

Master's Thesis

Master in Dynamics of Renewables-based Power Systems (DREAM)

Quadratic Curve Fitting Algorithm for PV Power Reserve Control

REPORT

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Resum

Aquest treball descriu el desenvolupament i les proves d'un algorisme de control de reserva de potència per al seu funcionament en una central fotovoltaica. El control de reserva de potència és un esquema de control que fa funcionar una font a una potència inferior a la màxima per tal de mantenir una reserva desplegable, en la majoria dels casos com a part de l'energia transitària proporcionada pels generadors síncrons virtuals en suport d'inèrcia per a la xarxa. El treball conté una revisió bibliogràfica que analitza les tècniques comunes per a aquest tipus de control fotovoltaic i la seva aplicació als convertidors que proporcionen inèrcia virtual. L'algorisme es basa a prendre un petit nombre de mostres dels punts de potència de reserva i a ajustar-les a una corba de funcionament quàdrica que en predueix la ubicació, la qual cosa permet operar amb una potència de reserva constant en condicions d'irradiància variables. L'algorisme es prova aïllat de la xarxa en una simulació de Simulink sota diferents condicions per avaluar-ne el rendiment i es compara amb una tècnica existent per al Control de Reserva de Potència. Finalment, l'algorisme es va provar en un experiment amb maquinari del mateix circuit que la simulació per avaluar-ne la viabilitat en condicions realistes. L'algorisme va demostrar ser eficaç en el seguiment de la potència de reserva, i ho va fer amb una precisió més alta que la tècnica comparada. Tots dos algorismes examinats van tenir dificultats per mantenir una reserva de potència constant en condicions de fluctuacions d'irradiància molt ràpides.

Resumen

Este trabajo analiza el desarrollo y las pruebas de un algoritmo de control de reserva de potencia para su aplicación en una planta fotovoltaica. El control de reserva de potencia es un esquema de control que hace funcionar una fuente a una potencia inferior a la máxima con el fin de mantener una reserva disponible, en la mayoría de los casos como parte de la energía transitoria proporcionada por los generadores síncronos virtuales para el apoyo a la inercia de la red. El trabajo incluye una revisión bibliográfica en la que se analizan las técnicas habituales para este tipo de control fotovoltaico y su aplicación a convertidores que proporcionan inercia virtual. El algoritmo se basa en tomar un pequeño número de muestras de puntos de potencia de reserva y ajustarlas a una curva de funcionamiento cuadrática que predice la ubicación de los demás, lo que permite el funcionamiento con una potencia de reserva constante en todas las condiciones de irradiación. El algoritmo se prueba de forma aislada de la red en una simulación de Simulink bajo diferentes condiciones para evaluar su rendimiento y se compara con una técnica existente para el control de reserva de potencia. Por último, el algoritmo se probó en un experimento de hardware con el mismo circuito que el de la simulación para evaluar su viabilidad en condiciones reales. El algoritmo demostró ser eficaz a la hora de realizar un seguimiento de la reserva de potencia, y lo hizo con mayor precisión que la técnica con la que se comparó. Ambos algoritmos analizados tuvieron dificultades para mantener una reserva de potencia constante en condiciones de fluctuaciones muy rápidas de la irradiancia.

Abstract

This work discusses the development and testing of a Power Reserve Control algorithm for operation in a PV plant. Power Reserve Control is a control scheme that operates a source at a power lower than the maximum in order to maintain a deployable reserve, in most cases as part of the transient energy provided by Virtual Synchronous Generators in inertia support for the grid. The work contains a literature review discussing the common techniques for this type of PV control and their application to converters providing virtual inertia. The algorithm is based on taking a small number of samples of reserve power points and fitting them to a quadratic operating curve that predicts the location of others, allowing operation at a constant reserve power across irradiance conditions. The algorithm is tested in isolation from the grid in a Simulink simulation under different conditions to test its performance and is compared to an existing technique for Power Reserve Control. Finally, the algorithm was tested in a hardware experiment of the same circuit as the simulation to assess the feasibility in realistic conditions. The algorithm proved to be effective at tracking reserve power, and did so at a higher accuracy than the compared technique. Both of the examined algorithms had difficulty in maintaining a constant power reserve in conditions of very fast irradiance fluctuations.

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List of Abbreviations

PRC Power Reserve Control

VSG Virtual Synchronous Generator

SC Supercapacitor

RES Renewable Energy Source

BESS Battery Energy Storage System

MPPT Maximum Power Point Tracking

RPPT Reserve Power Point Tracking

MPP Maximum Power Point

RPP Reserve Power Point

PV Photovoltaic / Power-Voltage

GFL Grid Following

PLL Phase-Locked Loop

GFM Grid Forming

DC Direct Current

AC Alternating Current

P&O Perturb & Observe

IV Current-Voltage

STC Standard Test Conditions

CPG Constant Power Generation

PI Proportional-Integral

GUI Graphics-User Interface

FPGA Field-Programmable Gate Assembly

1 Introduction

1.1 Motivation

The electrical grid is facing a transformation away from traditional fossil fuel-powered turbines towards renewable energy sources. As power electronics continuously represent a greater proportion of the sources supplying power to the grid, new challenges are being presented to maintaining grid stability. Principle among these is the frequency inertia that is inherently provided by synchronous machines [4]. During grid disturbances, the balance of energy supply and demand is maintained through speed adjustments in the generator's rotation. The rotating mass's physical inertia therefore also represents the grid frequency's inertia. The continuous replacement of synchronous machines with inverters results in a comparatively larger frequency disturbance for similar power imbalances, threatening grid stability and thereby slowing the transition to renewable resources [5]. Therefore, it is necessary to develop methods of creating frequency inertia using power electronic systems.

To accomplish this, a more popular field of investigation is Virtual Synchronous Generator (VSG) control, which incorporates the swing equation governing a rotating machine's inertial response into the control law of a grid-connected inverter [6]. Since Renewable Energy Sources (RES) are often operating at maximum power, many VSG implementations will include a form of energy storage to replicate the ever-present rotational kinetic energy of synchronous machines [2] [7]. However, these strategies have drawbacks. Battery Energy Storage Systems (BESS) add significant costs and complexity to projects, and the effectiveness of supercapacitors (SC) is limited to short time-scale power compensation. Thus, the authors of [2] proposed a coordinated strategy of both a supercapacitor and Power Reserve Control (PRC) replacing more traditional Maximum Power Point Tracking (MPPT) for a PV plant to supply the longer-term power for inertia regulation during under-frequency events. PRC maintains a sub-optimal operating point for the PV plant so that there is excess power available for deployment by the VSG inverter to replicate an inertial response [8].

The research group led by Professor Liu Jia at Xi'an Jiaotong University proposed a thesis topic for a novel strategy to implement PRC in a two-stage PV system based on sampling different Reserve Power Points (RPPs) and fitting them to a quadratic curve to operate along across irradiances. The proposal was created by a previous master's student and includes a theoretical outline for the strategy, a logical flowchart for implementing the algorithm, and a preliminary simulation of the algorithm employed in a utility-scale system.

1.2 Objectives

The objectives of this thesis project are to further develop the algorithm implementation from the proposal and verify its viability in a hardware experiment. The goal of the simulation and experiment will be to determine how well the proposed PRC algorithm can maintain a PV plant's operating power at a specific amount below the maximum power under varying irradiation conditions. The scope will be within the general theoretical framework proposed, which is a sampling-based algorithm that generates a quadratic curve of RPPs.

The first specific objective is to develop a new Simulink simulation. The original proposal was developed as a standalone proof of concept, and therefore requires several major revisions. The first is the implementation of the algorithm itself, which was designed as a MATLAB function block within the Simulink simulation. The function was created without attention to software

development best practices, most notably parameterization and modularity. Addressing these issues is crucial in order to ensure the maintainability of the software in the inevitable case that revisions become necessary in the future, as well as to ease its adaptation into C for the hardware experiment. Further explanation of the algorithm development is given in Section 3. The simulation case itself also needs to be adjusted to match the hardware available in the laboratory rather than a general utility-scale application.

Lastly, the PRC algorithm will be applied to a testbench experiment replicating the two-stage PV converter system described in Section 4. The proposal was intended to eventually be incorporated into the PRC-SC-VSG system in [2], so the experiment design will be a similar two-stage PV system. Through the results of the simulations and hardware application, this work aims to assess the proposed method's accuracy in maintaining an operating point at the given reserve power, as well as its resilience. Additionally, a more basic PRC strategy inspired by [1] will be implemented to provide a point of comparison for the proposed algorithm. Overall, this work aims to assess the PRC method's accuracy in maintaining a specific amount of reserve power and its resilience, which are quantified by simulation results and hardware testing in Sections 4 and 5. Through this assessment it also aims to identify further areas of work that could improve its applicability and performance. The work will be carried out as a mobility project hosted by the aforementioned research group in Xi'an, China. A layout of the tasks and an estimation of the necessary time for each is shown in Figure 1.

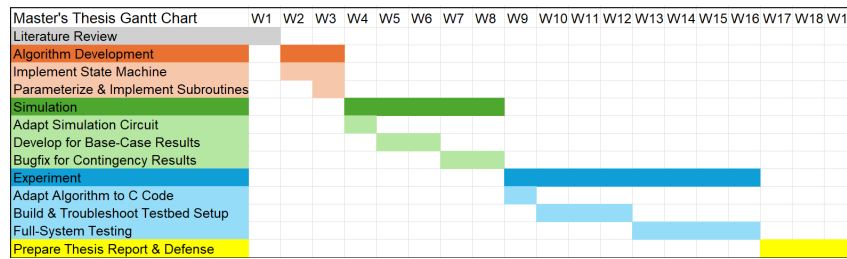


Figure 1: Gantt chart of thesis work

2 State of the Art

2.1 Virtual Synchronous Generator

The vast majority of currently deployed inverters for renewable sources are in the Grid Following (GFL) configuration. These inverters use a Phase-Locked Loop (PLL) to inject power exactly at the grid frequency. While it is possible to support primary frequency regulation by increasing the power setpoint during small disturbances, GFLs detract from total grid inertia [9]. Grid Forming (GFM) inverters instead set their own phase angle rather than locking to the grid's and can thus act as independent voltage sources. The method GFM use to determine this voltage frequency varies based on the goals of each particular use case. Frequency droop control is commonly employed to mimic synchronous machine governor action and allow for parallel operation analogous to conventional generators. However, this tactic is only useful for synchronization. To provide inertia support to the grid frequency, GFM have to specifically incorporate this property into the phase angle control [10]. This can be broadly referred to as Virtual Synchronous Generator (VSG). While other methods of creating virtual inertia such as Distributed Virtual Inertia [11] and Virtual Oscillator Control [12], VSGs are the most common and the intended application of this work.

Implementations of VSGs themselves vary widely, including various additional dynamics, damping terms, and droop controllers. The most basic and shared feature among these is the emulation of generator inertia by implementing the basic swing equation into the active power controller [13]. The swing equation represents the balance between electrical and mechanical power in a generator, shown in Equation 1:

$$J\omega_0 \frac{d\omega_r}{dt} + D(\omega_r - \omega_0) = P_m - P_e \quad (1)$$

where J , D , ω_0 and ω_r are the moment of inertia, damping coefficient, nominal rotor speed and current rotor speed, respectively. Since the moment of inertia and damping coefficient are control parameters rather than physical attributes of a generator, they can be modified as necessary for the desired $P - \omega$ droop and inertia time based on the relevant grid codes and considerations of the VSG's rated power [14].

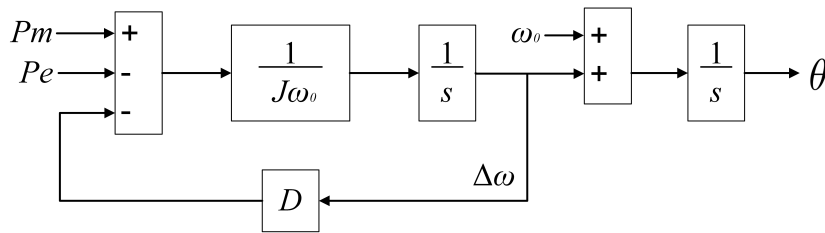


Figure 2: VSG swing equation control diagram

Figure 2 shows the swing equation applied to the active power controller of a VSG, with P_m being equivalent to the power reference, or power delivered from the DC side. In the event of a power imbalance caused by a shift in the grid angle, the inertia term replicates the slowing of a generator by modulating frequency based on the power difference. The larger the moment of inertia J is tuned, the smaller the change in frequency, just as a spinning generator will slow down less for the same power imbalance the larger it is. The response is also affected by the

damping or droop constant D . In a PLL-based inverter, the change in the grid angle would be rapidly accounted for by the PLL to maintain in-phase constant power output [15]. In the VSG case, the feedback droop term increases the power reference by a set proportion of the decrease in frequency, replicating governor action adding more mechanical power to a slowing rotor. This is the so-called frequency support energy that must be supplied from the inverter's DC side. The equivalent of the mechanical energy is a RES operating at maximum power instead of a spinning rotor, hence the need to artificially reserve energy on the DC side of VSGs. The PRC algorithm investigated in this work is also relevant to different VSG and inertia replication strategies, as they generally share the need to artificially reserve energy on the DC side.

2.2 Reserve Power

To provide reserve power for frequency inertia in PV systems, the goal is to maintain the power at a given point below the MPP. The main difficulty is determining what the MPP is while not operating at it, since in typical MPPT operation the power point is perpetually being approached based on ΔP and ΔV observations [16]. Thus, RPPT solutions typically include a method of determining the MPP as it varies with irradiance and temperature. Generally, the literature proposes three different methods to accomplish this: operating a MPPT string in parallel with the reserve power string, sampling and maintaining the RPP, or fitting an operation curve based on parameters or samples [17].

The first method avoids the ambiguity of estimating the MPP or fitting an operation curve for it by dedicating one more strings to constant operation at MPP. This MPP is then used to calculate an accurate reserve power point by simply to the rest of the strings at all times by simply subtracting the desired reserve power from the MPP [18]. The reserve strings will implement a power controller to track the calculated reserve power continuously. This tactic has the benefit of being highly accurate throughout the variations of the day, since the MPPT strings essentially act as a real-time measurement for the RPP. However, the assumption that the MPPT and RPPT strings are on the same curve breaks down in uneven shading conditions between the strings, and requires extra communication between the inverters to implement. This tactic is more practical in larger PV systems where leaving a few strings to MPPT does not have a significant effect on the potential reserve power.

Purely sampling-based PRC methods such as that proposed in [1] periodically sample the MPP to determine the power point to perturb to. The controller is divided between a MPPT mode and a Constant Power Generation (CPG) mode and uses a constant-voltage MPP method to increase the speed of sampling. The reserve power is subtracted from the sample MPP to calculate a RPP, which the controller maintains until a pre-determined resampling time. The modes can be switched more frequently to maintain a higher accuracy of the target reserve power during fluctuations or less frequently to maximize availability and reduce unnecessary samples during flatter irradiance. However, as shown in Figure 3, the actual size of the power reserve can vary widely as irradiance changes, which can affect the ability of the inverter to provide frequency inertia. This method is simple to implement and requires minimal adjustment to existing MPPT code, and is the one employed by [2]. As detailed further in Sections 4 and 5, this method will also be developed to provide a point of comparison with existing PRC methods for the proposed algorithm.

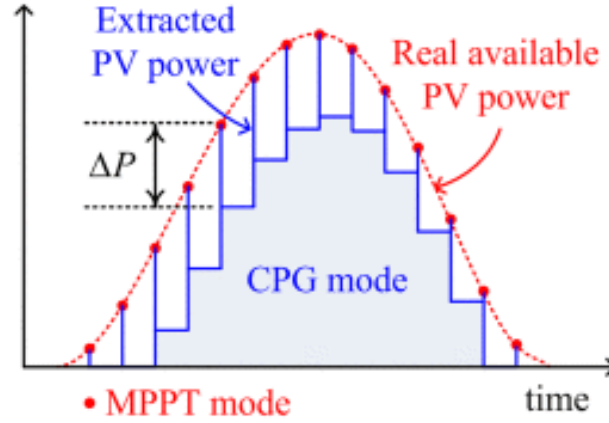


Figure 3: Constant Power Generation (CPG) method power reserves [1]

The most common solutions for PRC involve deriving the MPP based on the PV plant parameters and applying a curve fit to analytically derive the MPP or the entire PV curve to determine the reserve power operating point. They can be further distinguished into methods that derive a curve offline to be used during operation or fitting based on real-time operational data. The method proposed in [19] estimates the five parameters of the simple diode PV model (photocurrent I_{ph} , diode saturation current and ideality constant, shunt resistance R_{sh} , and series resistance R_s) based on datasheet parameters [20] to derive the maximum power for every irradiance G and temperature T pairing across the entire operating range. Finally, a regression to fit the matrix to the coefficients of $P_{mp}(G, T)$ gives the solution to Equation 2. This can be performed entirely offline and loaded into the PV controller before operation, and the experiment showed good accuracy in the results. The major drawback of this method and similar ones based on the simple diode model is that it requires irradiance and temperature measurements, which can add significant cost and complexity to the system.

$$P_{mp}(G, T) = d + a_1T + a_2T^2 + b_1G + b_2G^2 + cGT \quad (2)$$

The other family of curve-fitting methods generally take samples during operation and use them to fit a curve predicting the maximum power. [21] uses three operating points across the IV characteristic and fit a quadratic curve to predict the maximum power. [22] takes many samples at a small range around the RPP and employs a least-squares curve fitting method and a numerical derivation of the single diode model to create the IV characteristic curve in real time. These curve fitting techniques and other similar ones are generally effective but make tradeoffs between requiring high numbers of samples, large computational effort, and limited accuracy. The PRC algorithm developed in this work and discussed further in Section 3 attempts to increase the flexibility of a curve fit. This is accomplished by sampling points of the same amount of reserve power at different irradiances. This avoids constructing an entire characteristic curve as in most of the cited examples and instead fits a curve of exclusively RPPs along different irradiances.

Once a power reserve is established, deployment can be enabled with a communication link between the VSG inverter and the PV converter. The trigger condition is often a power increase that also triggers the VSG droop, and the PV converter controller is switched into MPPT. The extra power supplies the power reference spike caused by the frequency disturbance that would otherwise need to come from DC link capacitance or another energy source. A coordination of

supercapacitor and reserve PV energy is planned to provide a specific amount of energy during a power spike in [2], detailed in Figure 4.

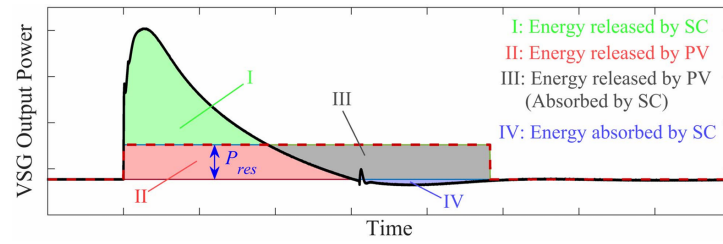


Figure 4: VSG energy sharing scheme [2]

Since the inertia and droop constants control how much energy is in the power transient for a given frequency disturbance, the VSG controller and the reserve power system can be designed in coordination to be able to provide a certain degree of inertia. The details of this are beyond the scope of this work but are explored further in [23].

3 PRC Algorithm

3.1 Theoretical Background

The most common circuit representation of a PV cell is the single-diode model, shown in Figure 5. It represents the light-induced current as a current source with a parallel diode, parallel resistance, and a series resistance. The parameters of this circuit are the photocurrent series resistance R_s , shunt resistance R_{sh} , diode ideality a , diode reverse saturation current I_0 , and photocurrent I_{ph} which are combined with the Shockley diode equation to produce Equations 3, 4, and 5:

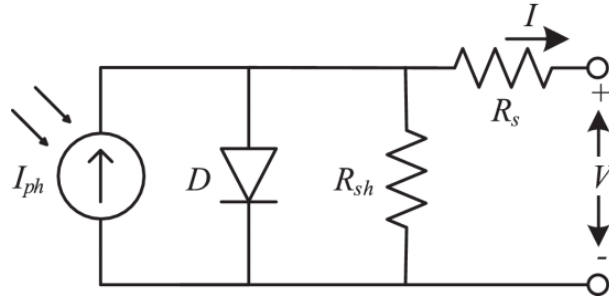


Figure 5: PV single-diode model

$$I = I_{ph} - I_0 \left(e^{\frac{V + R_s I}{V_T}} - 1 \right) - \frac{V + R_s I}{R_{sh}} \quad (3)$$

$$V_T = N_s T \frac{ak}{q} \quad (4)$$

$$I_{ph} = \frac{G}{G_{STC}} I_{sc} (1 + \alpha(T - T_{STC})) \quad (5)$$

Where k is Boltzmann's constant, q is the fundamental charge, N_s is the number of cells in series, V_T is the thermal voltage, G is irradiance, T is temperature, and STC is Standard Test Conditions. Since the equivalent circuit parameters are not typically given in manufacturer datasheets, there are various methods employed to estimate them based on the datasheet values or experimental results [24]. For the purposes of this work, the studied solar module will be the 1Soltech 1STH-215-P, a commonly studied panel with the diode ideality constant and resistances given by [25]. The other parameters are shown in Table 1. There will be 10 of them placed in series to approximate a realistic low-power application that can be realized with available laboratory equipment. To calculate the reverse saturation current, Equation 3 is solved at open circuit so that $I = I_{ph} = 0$ and $V = V_{oc}$ from the datasheet. Finally, by substituting the entire operating range for V , Equation 3 can be solved to produce the current I and power P characteristic. Figure 6 is created in this way from the 1Soltech parameters and shows a typical characteristic.

STC Power	213.15 W
Short-Circuit Current (I_{sc})	7.84 A
Open-Circuit Voltage (V_{oc})	36.3 V
MPP Current (I_{mp})	7.35 A
MPP Voltage (V_{mp})	29.0 V
# of Cells (N_s)	60
V_{oc} Temperature Coefficient	-0.361%/°C
I_{sc} Temperature Coefficient	0.102%/°C
Shunt Resistance (R_{sh})	313.399 Ω
Series Resistance (R_s)	0.394 Ω
Diode Ideality Constant	0.98117

Table 1: 1STH-215-P Specifications

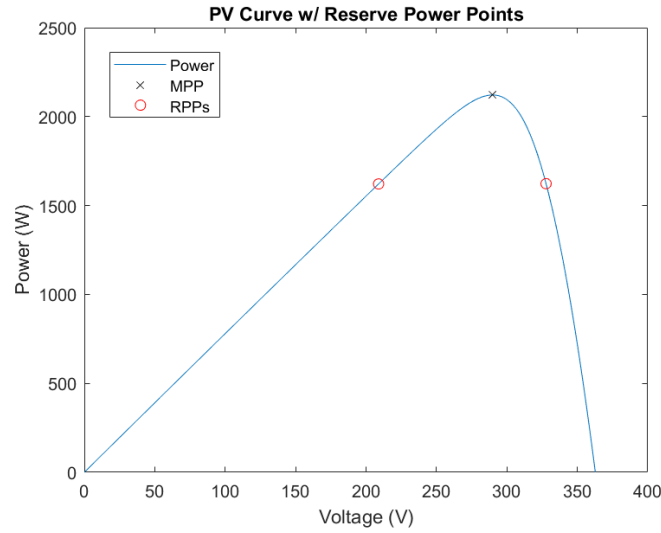


Figure 6: PV characteristic with salient power points

For every target reserve power P_{res} below the maximum P_{mp} , there are two potential operating points. The lower voltage side, or the left side, is generally a more linear region and thus may be easier to implement a power controller. Additionally, in a rapidly decreasing irradiance condition, the reserve power voltage will not become higher than the new open-circuit voltage, as is possible on the other side of the curve. However, the lower voltage also requires a much higher boost effort from the DC-DC converter. The right side region has a higher voltage that will lead to more efficient conversion. The steeper slope also will improve the speed of the controller, requiring a smaller voltage perturbation for the same change in power. For these reasons, the right-hand side is typically favored in the literature for PRC and is chosen in this work [26].

The general goal of a PRC algorithm is to determine the RPP that is as close as possible to a given ΔP from the MPP in any given irradiance or temperature conditions, while minimizing the extra measurements and samples needed.

$$P_{res} = P_{mp} - \Delta P \quad (6)$$

An assortment of P_{res} points across different irradiances at a constant temperature is shown in Figure 7. From the assortment of these points it is noticeable that they follow a continuous line. The main innovation of this work's PRC proposal is a fundamental shift in the curve-fitting approach to track RPPs based on this characteristic, rather than deriving it from the MPP. Estimation of the MPP in the PRC literature is only pursued in PRC applications as a means of determining the RPP. Deriving a curve along which all of a given PV plant's RPPs exist for a given reserve power would allow the actual PRC operation to be greatly simplified. Most of the cited PRC literature derives a PV curve at specific environmental conditions or requires sensors for them. A RPP curve across all of or most of the irradiance range would allow the power or voltage controller to simply perturb back towards the curve as the irradiance changes the operating point. To quantify the existence of such a curve, a least-squares curve fit is applied to all the RPPs in Figure 7, also commonly known as a quadratic regression. The high coefficient of determination $R^2 = 1.000$ indicates that the theoretical voltages of the same reserve power across different irradiances can be predicted accurately with a quadratic curve. However, determining this curve as shown would require samples from across the entire irradiance range before operation, similar to the requirements of the curve-fitting approaches in [21] and [27].

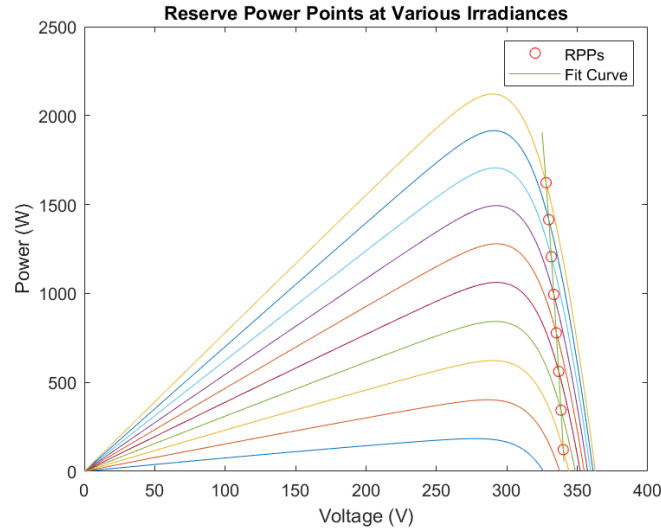


Figure 7: PV characteristics at 100 to 1000 W/m² irradiance with curve fit to RPPs

In order to minimize the number of samples taken to produce the reserve power curve, the PRC algorithm will fit three local samples $[V_{si}, P_{si}]$ to their characteristic quadratic by solving Equation 7. Taking three samples corresponding to the three unknowns A, B, C in the quadratic equation means that the resulting curve is guaranteed to pass through each sample, as shown in Figure 8. Using this approach of exact curve-fitting allows a parabola to be generated with the minimum number of samples, and is the chosen approach to contrast those in [22] and similar that can only fit after a high-number of samples. The tradeoff between number of samples and accuracy is common across applications and is discussed further below.

$$\begin{bmatrix} V_{s1}^2 & V_{s1} & 1 \\ V_{s2}^2 & V_{s2} & 1 \\ V_{s3}^2 & V_{s3} & 1 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} P_{s1} \\ P_{s2} \\ P_{s3} \end{bmatrix} \quad (7)$$

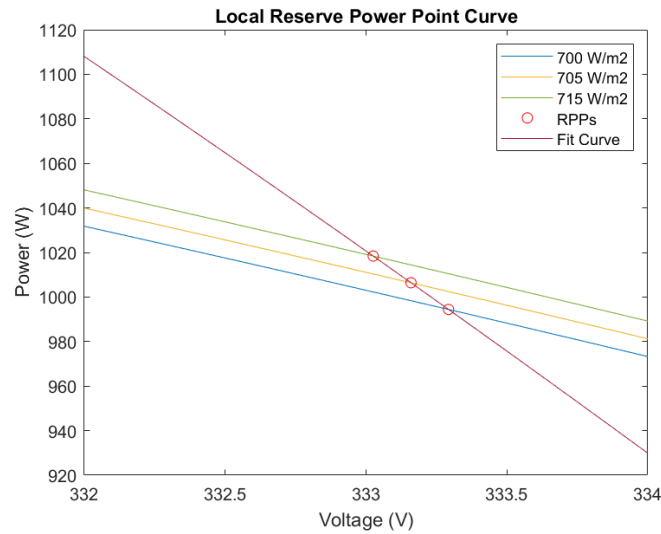


Figure 8: Curve fit between RPPs at small irradiance variation

In Figure 9, the curve fit from the small irradiance range (Figure 8) is shown in relation to the wider irradiance range. Checked against the larger set of reserve points, the curve has a lower determination coefficient $R^2 = 0.979$, which although is a significant decrease compared to the essentially perfect values of the more widely-sampled curve fit, would still be fairly accurate for predicting the entire range of RPPs. This phenomenon forms the intuition behind this PRC algorithm proposal: the reserve points of a panel across multiple irradiances form one tail of a parabola, which can be more or less accurately reconstructed with only a small sample of those irradiances. The algorithm will take three samples to construct this parabola presumably composed entirely of reserve power points. Then, as the irradiance changes and moves the power measurement off of the quadratic, the algorithm constantly adjusts the voltage back towards it. The result will be an oscillation around the reserve power point, similar to oscillation around the MPP during MPPT [16].

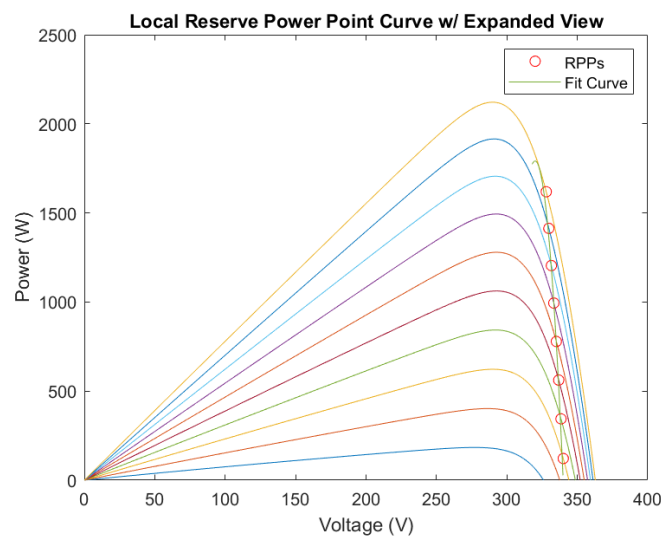


Figure 9: Curve fit from small irradiance variation

Finally, it is necessary to consider that this is the result derived from the single-diode model and perfect sampling. In reality, there will be sampling noise and slightly different conditions between the MPP measurement and the RPP sample. To illustrate the effect of this, random errors are added to the P_{res} value in the local irradiance samples and the curve fit repeated in Figures 10 and 11.

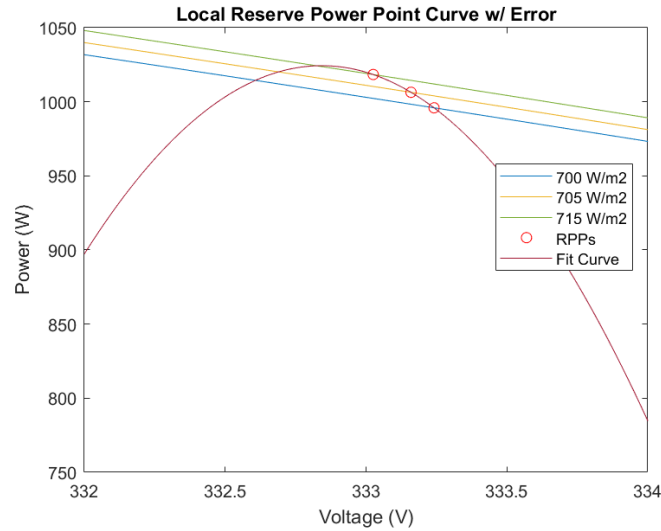


Figure 10: Curve fit from small irradiance variation with passable sampling error

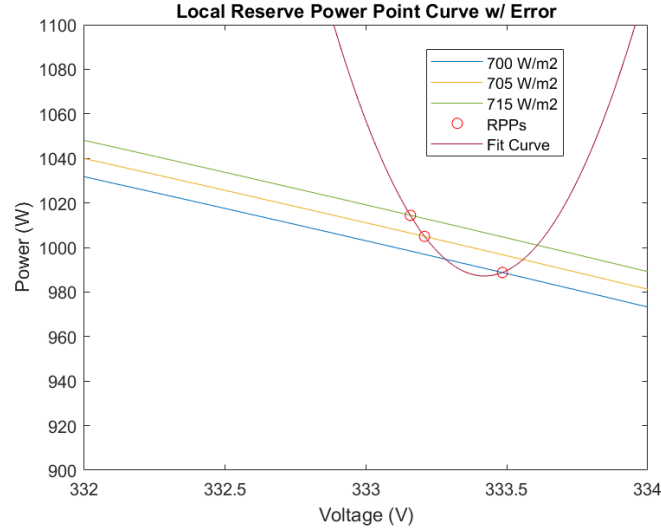


Figure 11: Curve fit from small irradiance variation with fatal sampling error

These error point curves make it clear that even a small imperfection in the RPP sample has a significant impact on the parabola and makes it unapplicable to the rest of the PV characteristic. Additionally, the apex of the curve in Figure 11 is at a lower voltage than the sample at the lowest irradiance. This means that as the algorithm attempts to decrease the RPP with decreasing irradiance, the reserve power curve will start increasing instead. This is an unsurprising complication, since it is common and expected that accuracy and number of samples are cor-

related. This tradeoff can be mitigated by being more specific about when samples are taken and which are accepted. These tactics are discussed in the next section concerning the specific implementation of the algorithm.

3.2 Implementation

The implementation of the algorithm is divided into three modes. The MPPT mode is used to determine the MPP, the relation to which is necessary for determining the RPP as in Equation 6. MPPT is also employed in conditions of low irradiance when the targeted reserve power is infeasible, and to deploy the reserve power. The second is the sampling mode, which determines what the reserve power is after the MPP is found and tracks to it. The sampling mode also includes a standby condition to ensure a minimum discrepancy between the power point in each sample. Last is the operation mode, which is the mode that performs the quadratic curve fit and operates along it. The logical flowchart describing the interaction of these modes is shown in Figure 12.

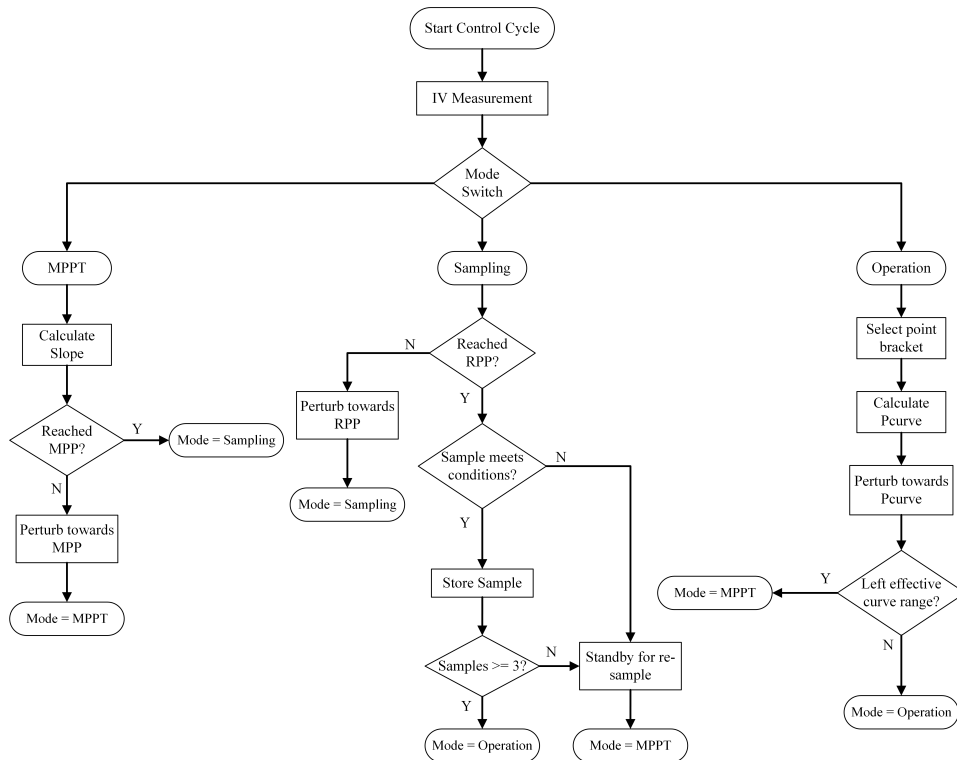


Figure 12: Algorithm logical flowchart

The most natural implementation strategy for this is a state machine, which is a programming abstraction that processes events and inputs and changes state based on the current one, in this case one of the three modes. In both C and MATLAB, a global struct can be used to store all the salient information as well as an enumeration representing the current mode. Then, a switch statement can direct the struct pointer to the corresponding subroutine based on the mode. This allows for a neat division of the sampling procedure and switch statement in the main function with each mode and helper in a separate subroutine. The flowchart represents a single control cycle, during each of which a new IV measurement is taken and a new reference

voltage is calculated. As seen in the previous section, small errors in the IV measurement can have a large impact on the quality of the curve fit. To mitigate this effect, an averaging period cycle is added during the IV measurement block, which takes the mean of a given period as the measurement fed to the control cycle. The averaging period is equal to the settling time of the voltage controller t_s and takes place after settling has finished, so the period of the control cycles is $2t_s$. The voltage controller is discussed further in Section 4.2.

3.2.1 MPPT Mode

The MPPT mode makes use of the commonly employed P&O algorithm for its simplicity and effectiveness. The controller of the DC-DC converter is thus a voltage controller, which is discussed further in Section 4. A fixed voltage step is applied to the controller and the change in power is observed. The process is repeated with the same sign in voltage step if it resulted in a power increase or the opposite sign if it decreased. This is a mature technology and works quite well, but requires an adaptation to be employed in this PRC algorithm. Since the principal purpose of this mode is to identify the exact MPP so the RPP can be calculated, it is insufficient to perpetually perturb towards the maximum power. It is necessary for the algorithm to actually identify when it has reached the MPP. To accomplish this, a feature of the incremental conductance MPPT technique is borrowed. Incremental conductance measures the slope of the PV characteristic $\frac{dV}{dP}$ to determine which way to perturb, and adjusts the step size based on how far away the MPP is [28]. Since the derivative of the PV curve will be zero as it changes from positive to negative, the MPP is findable as the only critical point of the characteristic. The P&O algorithm already tracks the change in power and voltage, so the slope is readily available with minimal adjustment. When $\frac{dP}{dV}$ is found to be below a set tolerance, the algorithm saves the maximum power and changes the mode to sampling.

3.2.2 Sampling Mode

The sampling mode is responsible for moving the operating point from the recently attained MPP to the RPP. The RPP is calculated using the stored MPP and the amount of reserve power, which is given as a parameter. A proportion of the error between the current power and the calculated target power is applied as a change to the voltage reference, similar to that employed in [29]. The right-hand side RPP exists on a fairly steep slope on the PV characteristic, meaning that a voltage step is prone to overshoot or oscillate around the target power. The proportional gain is thus chosen conservatively, according to the steepest slope $\frac{dP}{dV}$ at the highest irradiance. An integral gain is not employed for this reason, as well as the fact that the control effort and error are applied and calculated in discrete steps. It is not possible for the same step change in the voltage reference to be applied sequentially while the power error stays constant.

Once the RPP is acquired, the point must be checked before it can be added to the array of fit points. The most important is that each point in the array must have an increasing voltage with a decreasing power to match the general curve of the right-side reserve points, as shown in Figure 9.

$$V_{s1} < V_{s2} < V_{s3} \quad (8)$$

$$P_{s1} > P_{s2} > P_{s3} \quad (9)$$

Figure 13 shows a curve fit when the above equation is ignored in sampling. The fit will not even be applicable to the range just outside the samples, let alone the entire curve. The sampling

mode therefore includes a subroutine to sort the points into an array, and the conditions above are checked before the point is accepted.

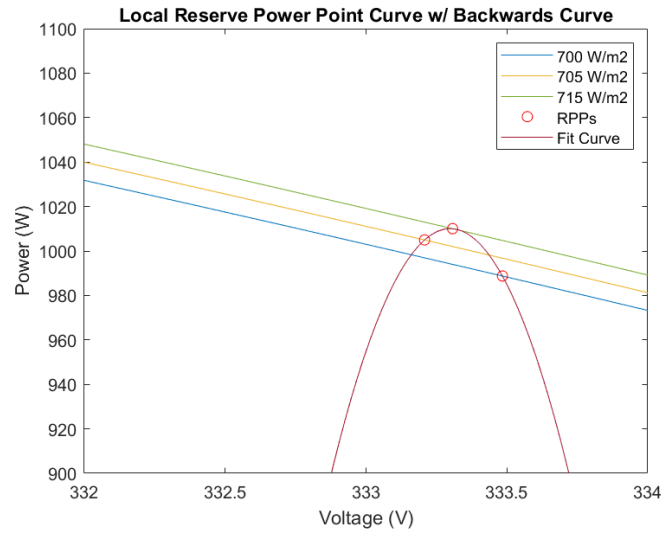


Figure 13: Curve fit from small irradiance variation with backwards samples

In order to minimize the chances of sorting failures and make the samples as useful as possible, the sampling mode also includes a standby condition to spread out the samples. The closer the samples are, the higher the proportion that sampling noise will be of the discrepancy between them. Thus, after each sample while there are still less than three, the algorithm maintains a constant voltage as an approximation of the fit curve in Figure 7 until the natural irradiance change has caused the power to drift a certain distance away from the previous sample point. This will ensure an approximately uniform distance between each sample and improve the accuracy of the resulting curves. The most recent RPP and MPP voltages are stored in memory and are jumped to immediately upon switching between the MPPT and sampling modes. This allows for much faster tracking of the desired points, since a new MPP is almost certainly closer to the previous MPP than the current RPP and vice versa. Once the third sample point has been acquired with the spacing and alignment conditions met, the algorithm changes to the last mode.

In the event that a sample taken after standing by still fails the sorting conditions, the algorithm will remove the point that breaks the condition in Equation 8 in order to accommodate the new point. Privileging the new points over previous ones in the array will prevent continuously oscillating sampling if a previous noisy sample is preventing a curve being fit at the new point.

3.2.3 Operation Mode

The operation mode is where the quadratic curve PRC takes place. Once entering from sampling mode, there will be an array of sampled points from which the algorithm chooses to construct the curve. The array is pre-sorted so that the voltage and power values are sequential, so the algorithm selects the point that is closest to the current operating power and the two adjacent points to form the curve. Once the curve is established, the voltage controller is perturbed towards the calculated P_{curve} rather than the MPP or RPP as in the previous two modes. This is visualized in Figure 14, with an exaggerated irradiance increase for clarity. The black X marks

the power drifting away from the power of the previous voltage reference as the irradiance changes. The operation mode will measure the power as being greater than the curve power and will adjust it along the new PV characteristic towards the RPP predicted by the curve.

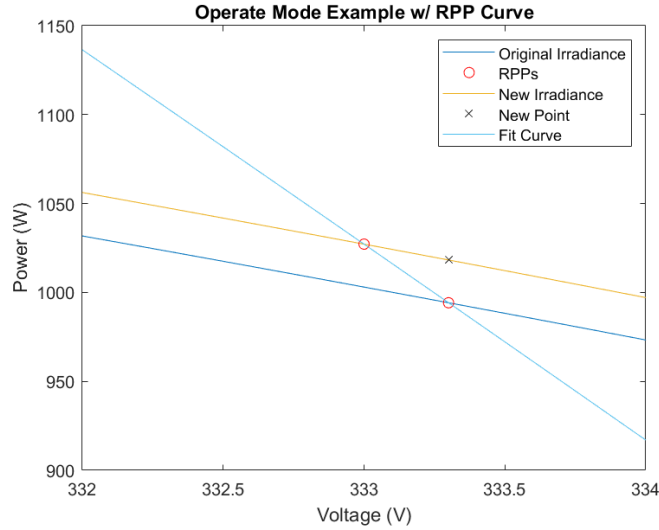


Figure 14: Operation mode curve-following visualization

However, this will only be as effective as the fit quadratic is accurate. Thus, the operation mode will monitor the operation point in relation to the fit curve as well as the sample points that have created it. In Figure 10 the expected shape of the fit curve is shown with the critical point just beyond the sample range. Since the parabola becomes invalid beyond this point, the algorithm will exit operation mode and resample the RPP when it is crossed. Since the quadratic coefficients are readily available, this critical voltage can be calculated using the derivative, shown in Equation 10

$$V_{crit} = \frac{-B}{2A} \quad (10)$$

The parabola is also assumed to require resampling outside of a range given as a parameter, which can be adjusted based on the accuracy of the samples and their resulting fit curves. The operation mode is also abandoned in very low irradiance conditions, which is detected by measuring the proportion of the RPP to the actual amount of reserve power, i.e. what the percent curtailment of the PV plant is.

4 Simulation

4.1 Circuit Design

The simulation will be designed in order to test the operation of the PRC algorithm in isolation. The reference paper [2] employs a two-stage system for the PV, which places a boost converter between the PV output and the DC link. This is a common approach and has the distinct advantage of increasing the DC link voltage, which allows the inverter to produce higher AC voltage. It also allows the separation of PV control from the inverter. The circuit tested will therefore be a boost converter with a PV input. The output will include the DC link capacitor and a voltage source to represent the management of DC link voltage. In reality this would be performed by another converter, such as in [2] or the inverter itself. For the scope of this work, however, it is sufficient to assume constant DC link voltage.

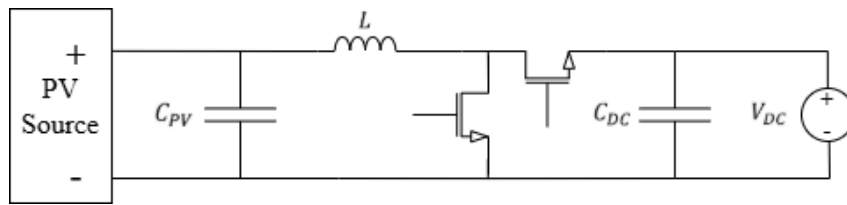


Figure 15: Ideal boost converter with input capacitance

The boost converter topology is shown in Figure 15. The circuit is realized with two transistors instead of a single one with a diode to allow the eventual hardware version to be built with an inverter arm. This converter and others switches those two transistors rapidly so that the stored energy in the inductor and capacitor raises the voltage at the output. Analysis of switching converters is done by splitting the circuit into its two switch states and averaging them using the duty cycle D , or $D' = 1 - D$. This averaging is used along with the characteristic relation for capacitor current and inductor voltage to write Equations 11 and 12, which will be the basis of the state-space model.

$$L \frac{di_L}{dt} = V_{PV} - D' V_{DC} \quad (11)$$

$$C_{PV} \frac{dV_{PV}}{dt} = i_{PV} - i_L \quad (12)$$

The inductor current in the boost converter ramps up and down during each switching cycle. To achieve steady-state operation with the input and output voltages constant, the energy released and by the inductor over one switching cycle must equal the energy absorbed. This is called the volt-second balance, and it is useful for analyzing converters in steady-state. The average inductor current over each switching cycle being constant means the left-hand side of Equation 11 can be set to zero to calculate how the output voltage is controlled by the duty cycle, shown in Equation 14.

$$0 = V_{PV} - D'V_{DC} \quad (13)$$

$$\frac{V_{DC}}{V_{PV}} = \frac{1}{1 - D} \quad (14)$$

For designing the converter itself, the biggest consideration is the inductor size, since it controls the inductor ripple Δi_L . To analyze the inductor ripple, we must abandon the volt-second balance approximation and analyze one half of the averaged equations to retain the ripple term. The expression for the inductor current ripple is calculated this way in Equation 16.

$$L \frac{di_L}{dt} = V_{PV} D \quad (15)$$

$$\Delta i_L = \frac{V_{PV} D T_s}{2L} \quad (16)$$

The circuit eventually implemented will be a low-power application based on the restrictions of available hardware. Thus, it will prove necessary to minimize the inductor current and ripple. The system parameters shown in Table 2 are chosen accordingly. With a DC link voltage of 400V, Equation 14 gives a duty cycle range of [0.1125, 0.3125] for $V_{PV} = [275, 355]$ based on Figure 7. Using a common switching frequency $f_s = 20\text{kHz}$, an inductor value of 3mH will keep the inductor ripple below 0.9167A in either direction from the average current and is conveniently available in the laboratory inventory. The capacitor sizes are also chosen for hardware availability purposes but are fairly large, resulting in negligible ($< 0.6\text{V}$) capacitor voltage ripple at steady-state.

DC Link Voltage	400 V
Switching Frequency	20 kHz
C_{PV}, C_{DC}	0.75 mF
Inductor	3 mH

Table 2: Boost converter parameters

4.2 Controller Design

The boost converter controller is used to set the operating point of the PV. Assuming a constant output voltage at the DC link, the duty cycle of the converter will therefore set the voltage of the input according to Equation 14. This algorithm makes use of a voltage controller for the PV operating point rather than a power controller, which is possible and employed in other works [30]. A voltage controller is suitable for this application because the P&O strategy used in MPPT is based on voltage perturbations. The PRC algorithm currently resamples and reperturbs towards the fit quadratic based on power discrepancies at constant voltage, meaning the algorithm provides a voltage reference regardless of the mode. The sample and operate mode could be reversed to use a power controller, but a voltage controller is preferred for the simplicity and convenience of using a single controller for all modes of the algorithm.

Section 3.2.2 describes the process of switching between MPPT and sampling modes where the previously stored MPP/RPP voltage reference is changed to immediately. This allows excellent

performance for finding these points but creates sharp current impulses with the rapid voltage change. In order to mitigate these, the boost converter will be controlled by two stages of PI controllers to allow dynamic saturation of the current as well as tracking to the voltage reference.

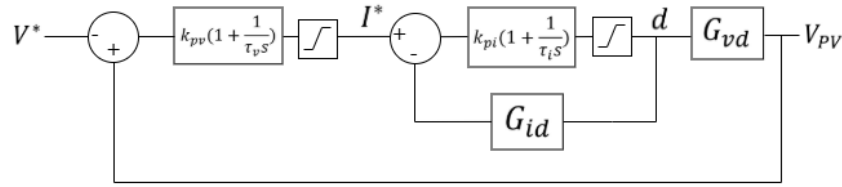


Figure 16: Boost-PV system block diagram

Figure 16 shows the block diagram of the entire boost converter and PV system. The inner current control loop will set the duty cycle of the converter, and the voltage reference from the PRC algorithm will control the current reference. Both the duty cycle and the current reference have saturation limits to respect the physical feasibility and practical limitations of the system, respectively. To determine the system transfer functions G_{vd} and G_{id} , a state-space is constructed from Equations 11 and 12.

$$\frac{d}{dt} \begin{bmatrix} i_L \\ V_{PV} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{L} \\ \frac{-1}{C_{PV}} & \frac{1}{r_{pvi}C_{PV}} \end{bmatrix} \begin{bmatrix} i_L \\ V_{PV} \end{bmatrix} + \begin{bmatrix} \frac{V_{DC}}{L} \\ 0 \end{bmatrix} d' \quad (17)$$

$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} i_L \\ V_{PV} \end{bmatrix} \quad (18)$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} i_L \\ V_{PV} \end{bmatrix} \quad (19)$$

Since V_{PV} is needed as a state in order to construct G_{vd} and G_{id} , the term r_{pvi} is introduced to relate the PV current term from Equation 12 to one in terms of the state, as done in [31]. This approximation only holds for regions of linear IV behavior, of which the studied array is divided into three in Figure 17. The resistances are 3.13k Ω , 39.37 Ω , and 7.46 Ω for the high current, maximum power, and high voltage regions, respectively. Since the operation of the PRC algorithm will be largely in the high voltage region, the controller will be designed assuming $r_{pvi} = 7.46\Omega$ and then the performance will be re-tested with a higher resistance to ensure that the response is still acceptable. Because the inductor resistance R_L is neglected, the r_{pvi} term is only present in the damping constant ζ as shown in Equation 23, so the extent of adverse effects should be limited to slightly increased voltage oscillation at lower voltages.

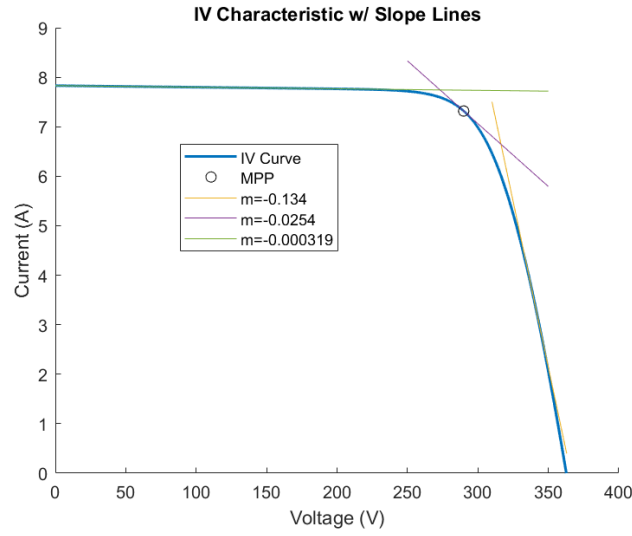


Figure 17: IV curve with approximated linear regions

With all the parameters accounted for, the transfer functions G_{vd} and G_{id} can be extracted with the property 20 and by switching the 1 and 0 in the C matrix as shown in Equations 18 and 19.

$$H(s) = C(sI - A)^{-1}B + D \quad (20)$$

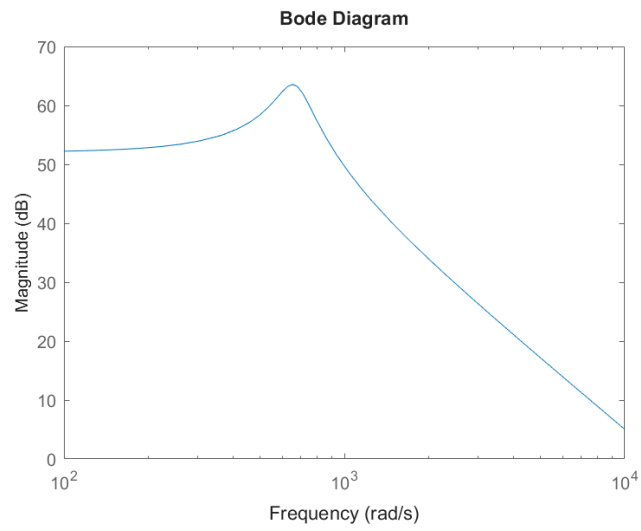
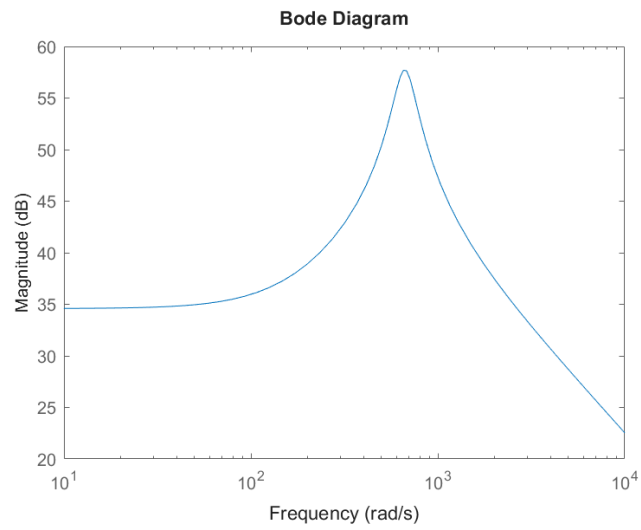
$$G_{vd} = \frac{K}{s^2 + 2\zeta\omega s + \omega^2} \quad (21)$$

$$K = \frac{V_{DC}}{LC_{PV}} \quad (22)$$

$$\zeta = \frac{1}{2r_{pvi}C_{PV}\omega} \quad (23)$$

$$\omega = \sqrt{\frac{1}{LC_{PV}}} \quad (24)$$

The Bode magnitude plots for the system transfer functions are shown in Figures 18 and 19. Both the responses have significant resonance peaks at $\omega = \frac{2000}{3} \frac{\text{rad}}{\text{s}}$. The transfer function of a PI controller is shown in the block diagram in Figure 16, it contains a zero at $\omega_0 = \frac{1}{\tau}$ and a left-hand plane pole. The maximum magnitude damping of the PI controller is after the zero frequency when the pole and zero have cancelled. The time constant of the outer voltage controller is thus selected to place the zero before the system resonance peaks for maximum damping. The current controller's time constant is a decade faster in order to prevent interference between the inner and outer loops.

Figure 18: Bode magnitude plot of G_{vd} Figure 19: Bode magnitude plot of G_{id}

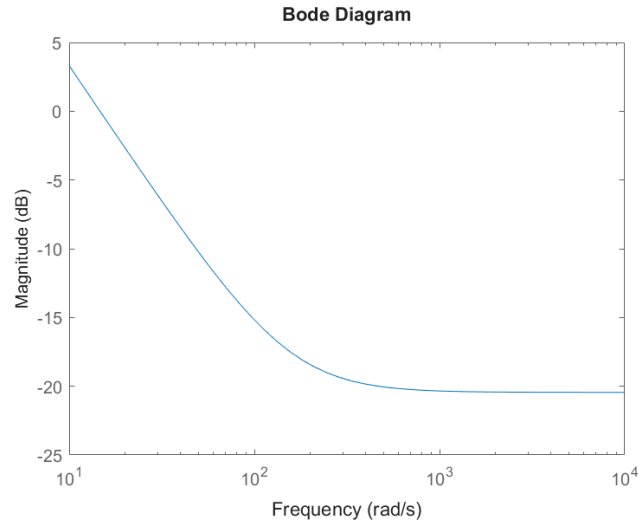


Figure 20: Bode magnitude plot of voltage PI

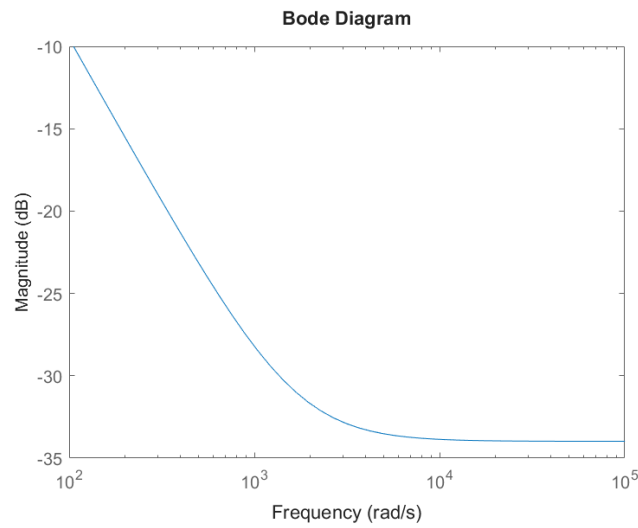


Figure 21: Bode magnitude plot of current PI

The Bode magnitude plots of the PI controllers show the damping at the resonant peaks of the system responses. After setting the time constants, the proportional gains are set to balance between a fast time response and minimal overshoot. The transfer function of the entire closed-loop system is assessed for stability in Figure 22, and the results of a voltage reference step on the full system from 300 to 350V are shown in Figure 23.

τ_v	0.006
τ_p	0.0006
k_{pv}	0.1
k_{pi}	0.1

Table 3: Boost PI controller parameters

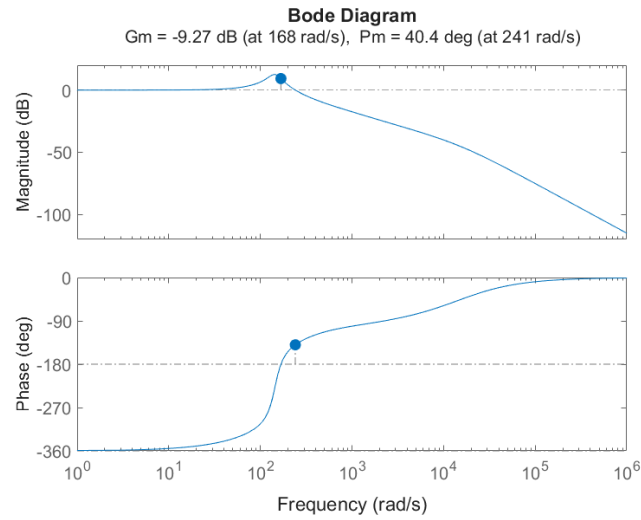


Figure 22: Bode plot of closed-loop system with phase and gain margins

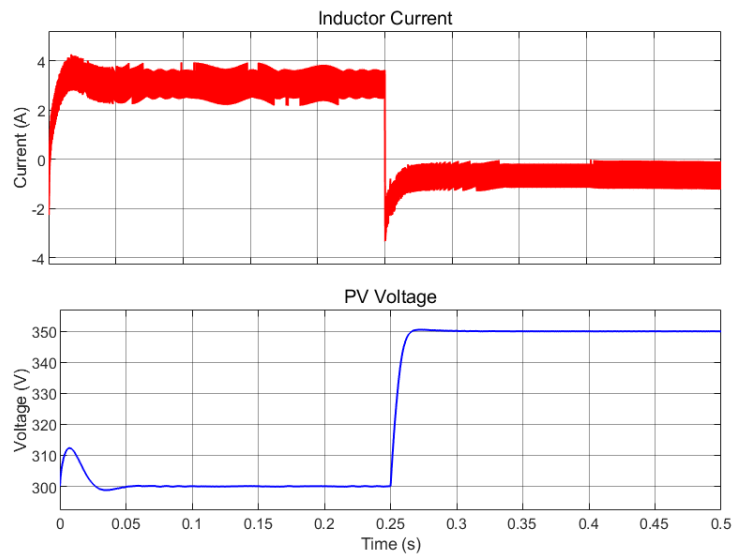


Figure 23: Closed-loop controller step response from 300 to 350V

Both results for the controller are indicative of good performance. The total inductor current ripple is about 1.8A at the highest, which is double the half-ripple that was predicted in Equation 16. The gain and phase margins are within healthy ranges for stability, and the step response is fast with a very small overshoot. This will be the expected behavior of the controller for the majority of the operating range. However, the assumption of a low r_{pvi} term becomes less valid as the voltage decreases. To test the performance outside the intended range, the step change is repeated at a much lower voltage to replicate the high-resistance region on the other side of the MPP.

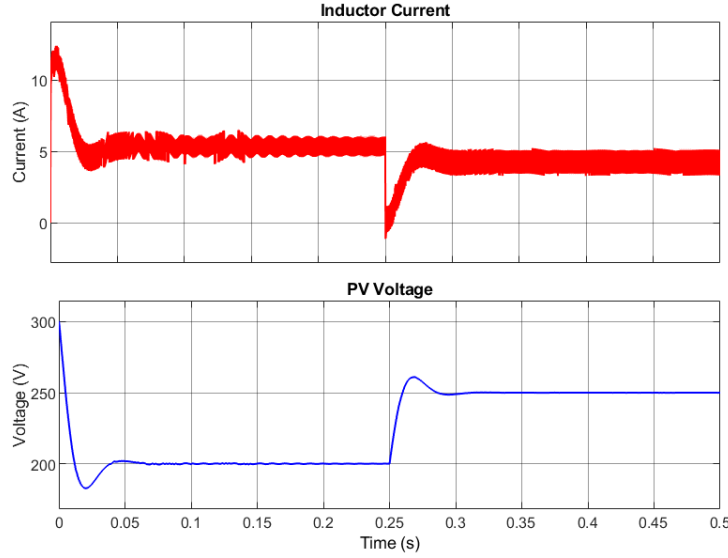


Figure 24: Closed-loop controller step response from 200 to 250V

Figure 24 shows a higher overshoot as expected, since the actual r_{pvi} is much higher in this range and the damping is therefore lower. Despite 200-250V being the low range of feasible operation and outside of the expected range of MPPs and right hand RPPs, the oscillations are still not extreme. The controller design with the low r_{pvi} assumption is therefore accepted. Between Figures 23 and 24, the settling time of the voltage controller is estimated to be $t_s = 0.05s$. Leaving another 0.05s to take an averaged measurement will result in 10 control cycles per second for the algorithm as a whole.

4.3 Methodology

To test the performance of the algorithm, irradiance profiles are given to the input of the Simulink PV block and calculated according to the panel parameters. The objective is to assess how well the algorithm maintains reserve power in different conditions. For the best approximation of real-world environmental conditions, irradiance data taken at one-second intervals from a cloudy day in Oahu, USA is used [3]. Temperature data is also applied from the same day and location but at a much larger 30 minute interval, since the high fidelity irradiance project did not take temperature measurements [32]. To measure the success of the RPP tracking, a second PV array is operated on a separate, identical circuit at the same environmental conditions but with a MPPT controller. Thus, the RPP at each time can be obtained by subtracting the reserve power ΔP from the power of the parallel PV array. Where applicable, this is used to calculate an error e for the PRC algorithm according to Equation 25.

$$e(t) = \frac{|P_{PRC}(t) - (P_{mp}(t) - \Delta P)|}{P_{mp}(t) - \Delta P} \quad (25)$$

There are several irradiance profiles that have been used in the simulations and subsequent experiments, including three core profiles that cover the range of functionality testing. First, a basic piecewise ramp irradiance that has a much higher slope than natural irradiances was the first test for each iteration to assess the essential operation faster than with real-time irradiance,

since the simulation must have covered a large enough irradiance difference to produce three quality samples before operation mode turns on. This profile quickly covers a large enough region to enter operation mode without a costly simulation. Second is a dramatic plunge in irradiance from mid-morning brightness to thick cloud cover followed by a rapid recovery. This profile contains the largest sudden drop in irradiance from the selected day and also plunges to a very low irradiance. This allows testing of the reserve power tracking during very fast irradiance shifts, as well as the low-irradiance condition that switches to MPPT when the targeted reserve power becomes too large a proportion of the available power. There is also a milder irradiance drop that will not trigger the switch to MPPT to observe the PRC operation's response to irradiance fluctuation within a higher power range.

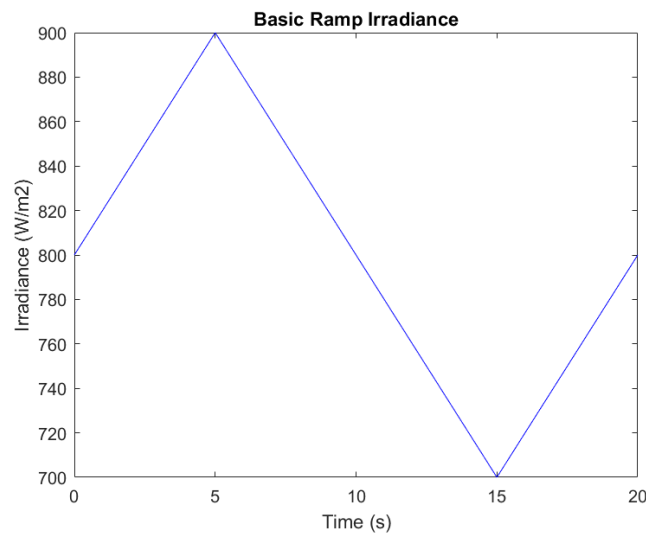


Figure 25: Basic ramp test irradiance profile

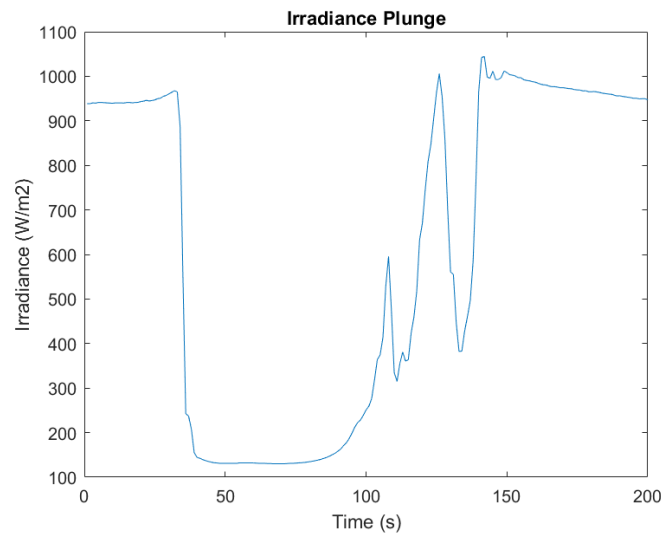


Figure 26: Profile of drastic irradiance drop

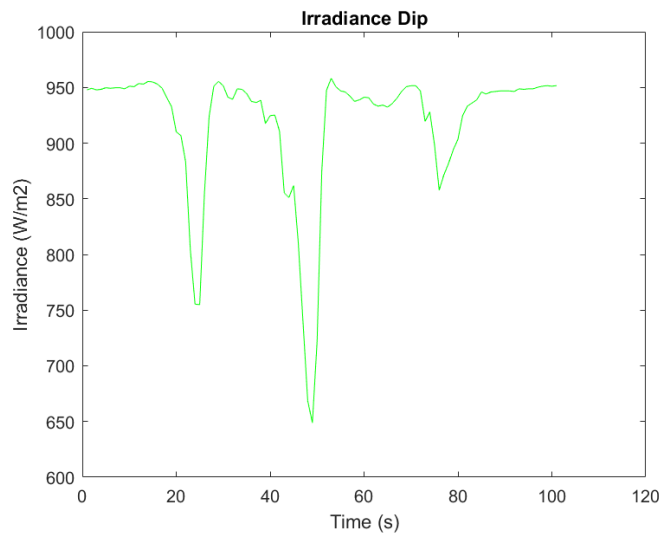


Figure 27: Profile of mild irradiance dip

These profiles will test the PRC algorithm's primary function of maintaining a constant power reserve. The other function that will be tested is actually deploying that power. According to [2], this is accomplished by switching from PRC mode to MPPT mode, thereby suddenly making the extra power available through the DC link. This is also the strategy in [1], so it is taken to be typical. Since the inverter control is outside the scope of this work, the command to deploy the reserve power will be simulated by a step response to a boolean in the Simulink function block.

4.4 Results

The first test result is of a ramp with a highly accelerated irradiance slope of 20W/m^2 per second, which is more than one hundred times faster than the rising mid-morning sun increases irradiance. This is done out of a practical concern for quickly assessing the viability of each algorithm iteration. Any errors in logic or poorly tuned parameters will make themselves readily apparent without an extended simulation. The PRC algorithm under this ramp irradiance is shown in Figure 28.

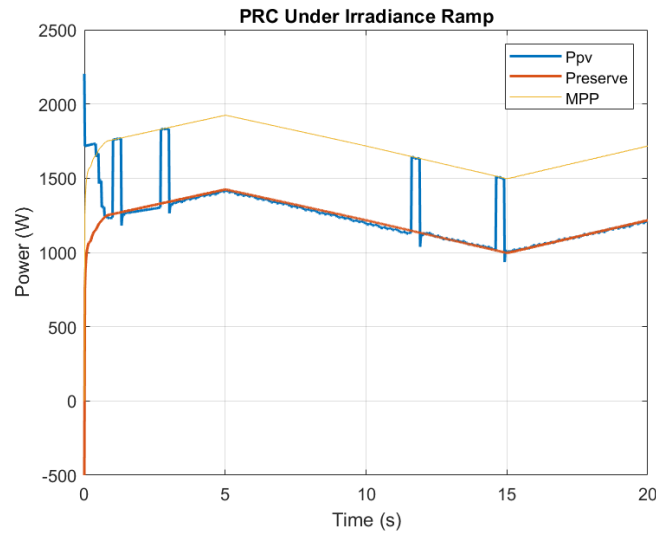


Figure 28: PRC algorithm under test ramp profile

This result shows the characteristic behavior of the PRC algorithm. The line labelled 'MPP' is the parallel PV array constantly tracking the maximum power, and the label 'Preserve' indicates this maximum power subtracted by the set reserve power, which is 500W. The line 'Ppv' shows the actual output of the PV plant controlled by the PRC algorithm. The first few seconds show the sampling procedure, where the quadratic curve is still being built. The impulses are switches to MPPT and back to sampling mode. The new MPP is used to track the RPP, and once it's found the sampling mode holds at a constant voltage. A slight drift away from the reserve power line is apparent in this standby period since there is no curve yet to follow it accurately. After the third sample is taken before 5s, the curve is created and the controller starts the operate mode, during which it is clear that the reserve power is tracked quite well. The average error between this reserve power line and the PRC output power is 4.48%. While not entirely meaningful on its own, the error figure is useful for comparing different algorithms. A basic CPG PRC implementation resembling that proposed in [1] is shown in Figure 29.

The principle of the CPG PRC is much simpler; it simply holds constant power until a set time has passed and then resamples to track a new RPP. The resample time parameter used in [2] is 6 seconds. A lower resampling time improves the reserve power tracking at the cost of increased samples, while a higher resampling time will have a higher accuracy in flat irradiance conditions at the cost of lower accuracy in changing irradiance conditions. The CPG PRC algorithm is tested for comparison to the proposed algorithm only, so a detailed optimization of this parameter is considered out of the scope of this work.

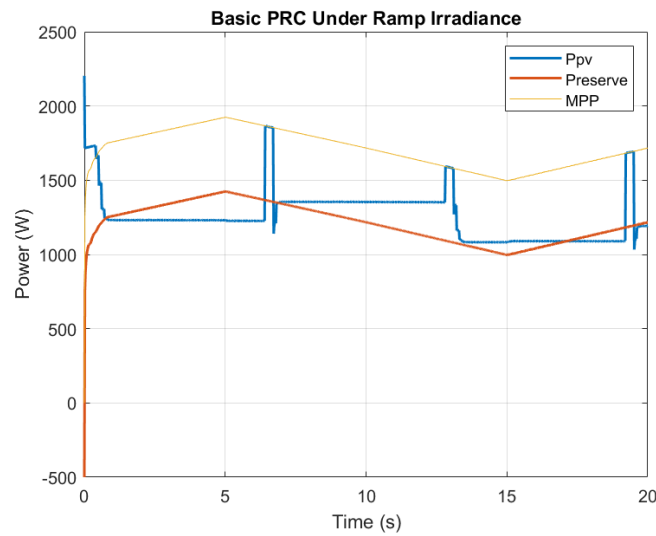


Figure 29: CPG PRC under test ramp profile

The average error of this profile is much higher at 10.38%. This is entirely expected, since the CPG algorithm is not attempting to continuously track the RPP with irradiance changes and is just an estimation. The exaggerated irradiance increase also exaggerates the error, since the CPG algorithm is based on the time elapsed. However, it does illustrate the issue of the CPG implementations not maintaining consistent reserve power. An unpredictable reserve power available to the system makes it more difficult to accurately plan the inertia response as in Figure 4 and further suggests the quadratic-fitting algorithm's goal of tracking a constant power reserve is plausible.

To explore its robustness further, the plunge irradiance profile is applied in Figure 30. The low irradiance condition will be set to trigger at 50% curtailment of the PV plant.

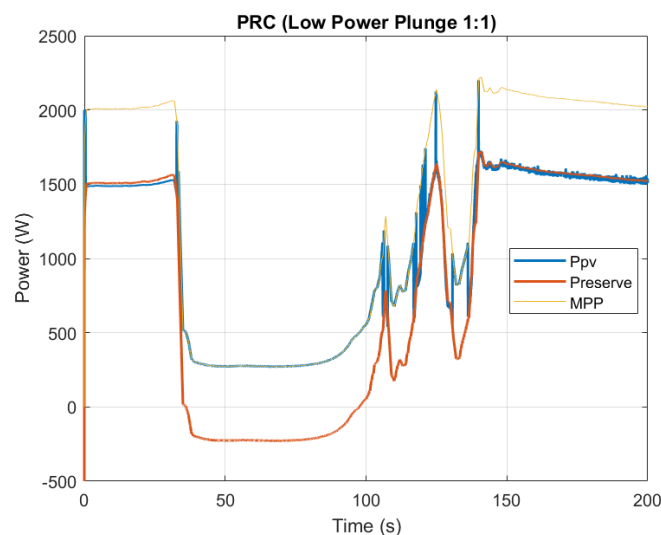


Figure 30: PRC algorithm under plunge irradiance profile

The first section preceding the plunge shows a small error in the reserve tracking during standby constant voltage mode. This represents a limitation of each of the simulations, which is that each simulation starts with an empty sample array, when in a real application it would likely have previous samples. In order to assess typical daytime behaviors, it is necessary to start at a higher irradiance and an empty sample array. The irradiance plunge itself shows a very timely switch without oscillation into the MPPT mode and steady operation in the trough while the target reserve power remains higher than the available power. The return to normal PRC operation is allowed above twice the power reserve of 500W, which explains the MPP spike above 1000W triggering extra sampling just after $t = 100$ s. The return above the curtailment limit at about $t = 125$ s again shows oscillation as the power threshold for resampling is repeatedly exceeded by the irradiance spike. The repeated sampling behavior is caused by the steep irradiance changing the IV characteristic relative to the speed of the control cycles. By the time a MPP is taken and its corresponding RPP tracked, the increasing irradiance has shifted the curve so that the RPP sample taken is at a much different irradiance than the MPP used to calculate it. If this difference is too large, the sample is rejected by the rest of the array and the process repeats. After the initial sample bout and before the sample at the peak, however, the RPPT is accurate even in the spike condition. This is also exemplified in the power decrease after the MPP spike at 2000W, during which there is accurate RPPT all the way down to the low irradiance condition. Finally, after the clouds finish passing and the irradiance settles at $t = 150$ s, the operation mode tracks the reserve power cleanly. These results suggest that although rapid irradiance drops into unsampled regions will create oscillatory impulses between MPPs and RPPs, the quadratic curve-fitting approach works well to track the irradiance drops through already sampled regions. Also, it is noted that the error calculation and accuracy comparison of the RPPT is not done for this irradiance profile due to the low-irradiance condition.

Finally, a milder irradiance dip profile is shown in Figure 31. These dips will not cause the array to exceed maximum curtailment. This allows for an accurate error figure to be calculated and to observe the RPP tracking during a dip without changing modes to MPPT.

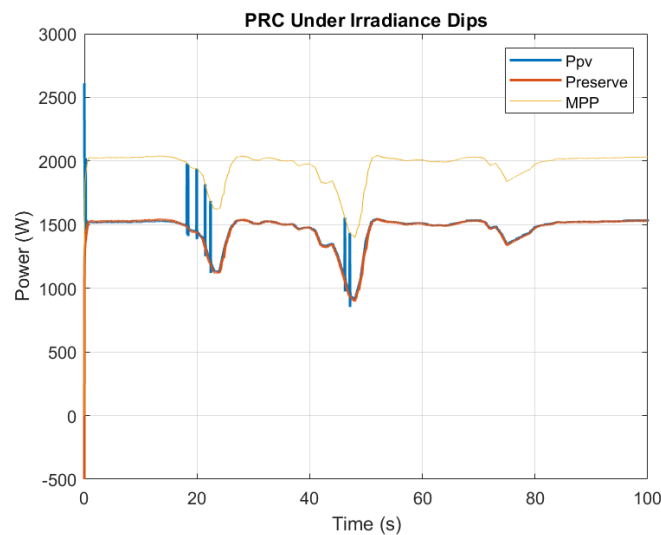


Figure 31: PRC algorithm under dip irradiance profile

The milder irradiance profile appears to have a huge effect on the tracking, with the RPPT only

having 0.61% error. The slower irradiance changes compared to the ramp and the plunge profiles allow for much more accurate tracking. The steepest irradiance drops in this profile give only a 140W/s drop in maximum power, compared to 338.7W/s in the plunge profile. The result is that the control cycles are quick enough to locate the MPP and the RPP before the IV characteristic shifts too far and the point becomes invalid. After $t = 50$ s, the irradiance remain within the sampled region and the reserve power tracking performs with high accuracy. This simulation is contrasted with that of the CPG PRC algorithm, which is given the same irradiance profile in Figure 32.

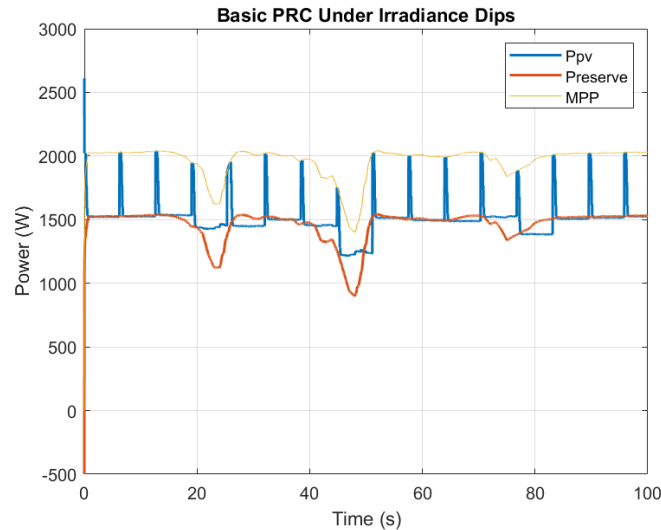


Figure 32: CPG PRC under dip irradiance profile

This simulation produced a much higher average error of 6.13%, which shows the weakness of the CPG PRC implementation. While the slower-changing irradiance regions show a drastic improvement to its reserve power tracking compared to Figure 29, the reliance on constant power means it cannot detect any irradiance changes and drifts off during the fluctuations. The PRC algorithm therefore represents a significant improvement to reserve power tracking accuracy.

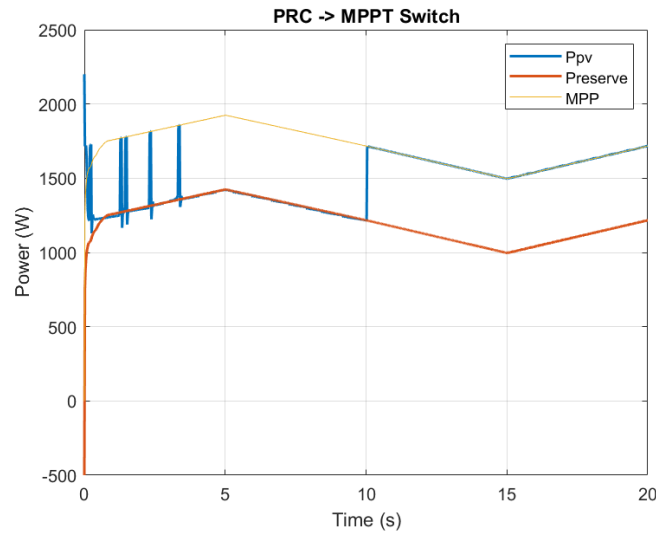


Figure 33: Power reserve deployment

Finally, the deployment of reserve power is tested on the ramp irradiance profile in Figure 33, with a step at $t = 10$ s to replicate a command from the inverter to release the power reserve. The results show a seamless switch to MPPT. Considering that an instantaneous switch to MPPT is already a built-in feature of the PRC algorithm, the successful deployment of reserve power on command is entirely expected.

Profile	Proposed PRC error	CPG PRC error
High-Speed Ramp	4.48%	10.38%
Plunge	N\A	N\A
Dip	0.61%	6.13%

Table 4: Simulation RPPT error results

Overall, the simulations have proved the feasibility of the proposed PRC algorithm in tracking reserve power in different conditions. The most difficult challenges for it appear to be in rapid irradiance changes through unsampled regions, when the changing MPP rivals the speed that new RPPs can be sampled. In general, the simulations have also displayed the superior accuracy of the curve-fitting method compared to the CPG one. The next section will seek to replicate this results on the hardware test bench.

5 Hardware Experiment

5.1 Design

The test bench circuit will be designed exactly as in the simulation, the schematic of which is shown in Figure 15. The second switch of the boost converter is implemented as such instead of a diode, which is also commonly used in switching converters. This is done because the most readily available switching hardware is an inverter, and a two-switch realization allows the use of one of the inverter branches to be turned into a DC-DC converter. The DC voltage and ground connections will remain the same, with the node in between the switches that would normally be one phase of an AC input will instead be connected to the PV system through the inductor. This is accommodated in the control software by inverting the duty cycle such that $D = 1 - D$, since the duty cycle conventionally refers to the positive transistor's duty cycle in the inverter controller. The controller will be the MywayPE4-PEV system, which incorporates a custom specialized C library with a data monitoring GUI and FPGA boards to control the converter and take measurements. The DC voltage source on the DC link bus is replicated with a IT6006C bi-directional DC source set to constant voltage mode, and the PV is replicated with another DC source of the same model controlled by the SAS1000L PV emulation software. The emulation software uses a different model than the single-diode model described in Section 3.1 and created a much larger drop in V_{OC} and V_{MP} than predicted in the simulations.

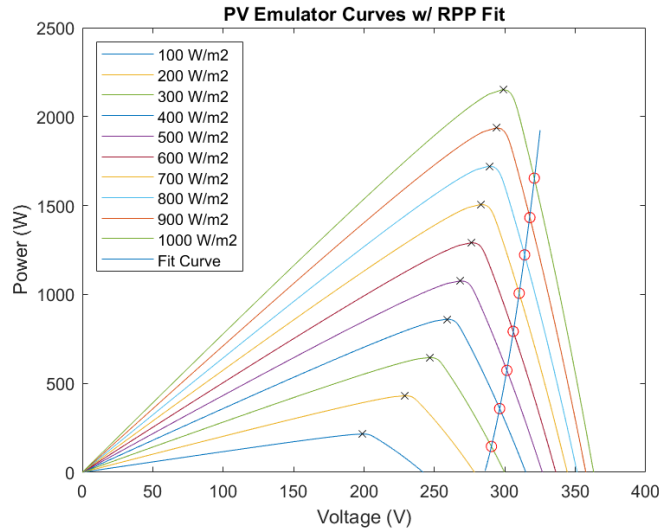


Figure 34: SAS1000L PV emulation characteristic at 100 to 1000 W/m² irradiance

The characteristic in Figure 34 is generated by the software based on the datasheet parameters. There are configuration options available for the temperature coefficients and fill factor, but the V_{OC} and V_{mp} calculations themselves are proprietary and not accessible to the user. The consequences of this are minor, since the quadratic nature of the RPPs is preserved ($R^2 = 0.999$). The only adjustment necessary is that the sequentiality constraint defined in Equation 8 must be reversed, since the RPPs for a given power now also have a decreasing voltage as the irradiance decreases. Other than this adjustment, there are no differences between the MATLAB PRC algorithm implementation and the embedded controller C implementation. An adaptation of the CPG PRC algorithm is also made into C for comparison. Additionally, in order to

quantify the error of the RPP tracking, a measure of the MPP at all times is needed. However, due to hardware constraints in the laboratory it is not possible to operate a second PV-boost circuit as in the simulations. The PV emulation software provides salient measurements during execution, including the MPP. The Field Programmable Gate Assembly (FPGA) controlled by the MywayPE4 unit tracks the PV current, PV voltage, DC voltage, duty cycle, and any intra-algorithm variables at a higher fidelity than the PV emulator's. This data and the PV emulator's are time-aligned in the post analysis to calculate the actual RPP at each time, allowing for an error calculation analogous to the simulation results. Thus, the general methodology of the experiment will be to apply various irradiance profiles to the PV emulator, measure the powers created by the two PRC algorithm, and compare these values to the actual RPP calculated from the emulator.

5.2 Results

Since the hardware controller is capable of executing the control algorithm in real time, the simple test ramp irradiance profile used in simulation is replaced with a typical sunny increasing irradiance of about 4W/m^2 per minute over a much longer period of time to capture the process of acquiring the first samples in a more realistic scenario, shown in Figure 35. The results of applying this profile to the proposed PRC algorithm and the CPG PRC algorithm are shown in Figures 36 and 37, respectively.

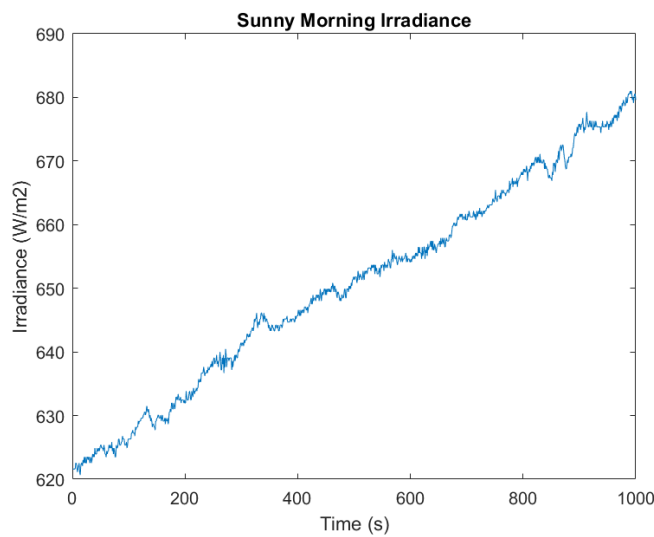


Figure 35: Typical sunny irradiance profile

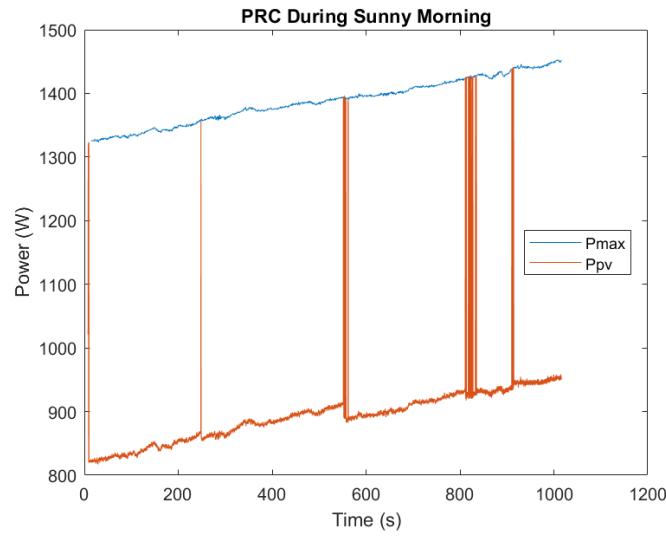


Figure 36: Proposed PRC algorithm under sunny irradiance profile

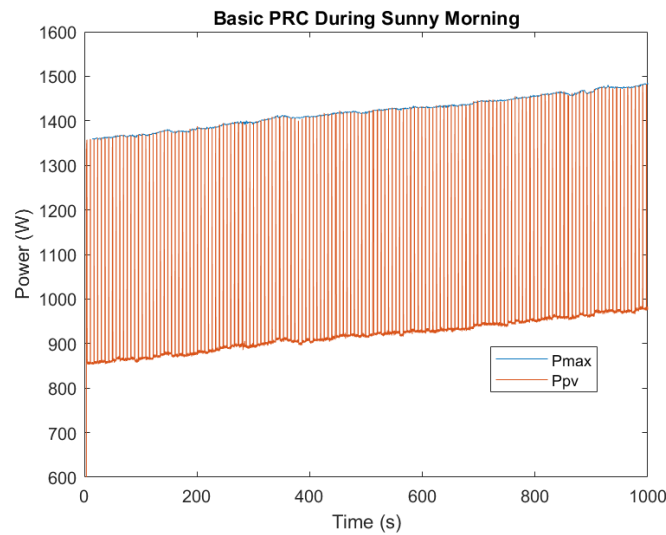


Figure 37: CPG PRC algorithm under sunny irradiance profile

The results are largely what is expected based on the test ramp simulations. The irradiance is no longer a perfect ramp but instead has oscillations as it increases, which is expected for having 1-second measurement intervals from a real sensor system. This, combined with the natural measurement noise results in more sample rejections than in the simulation, especially visible just past $t = 800$ s in Figure 36. After the third sample is captured, however, the operation mode does track the reserve power accurately. The average error is 1.37%, compared to 7.20% from the CPG PRC algorithm. The resample time of 6 seconds is clearly excessive in Figure 37, representing the tradeoff inherent to the algorithm. Higher sampling rates allow for the constant power setpoint to be more accurate, but the sampling is itself a deviation from the RPP. To compare with a version better tuned for this profile, a resample time of 60 seconds is implemented in Figure 38.

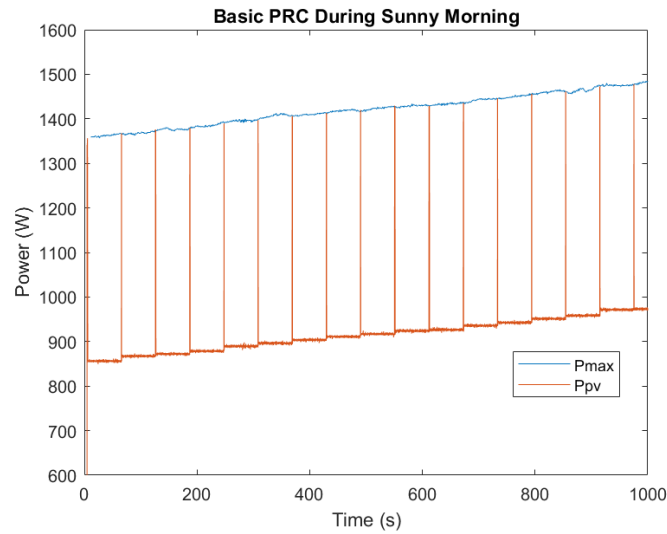


Figure 38: CPG PRC algorithm with extended resample time under sunny irradiance profile

Increasing the resample time of the CPG algorithm gives a greatly reduced error of 2.68%. The low variation in irradiance was causing the excessive sampling to reduce the accuracy more than the updated RPPs can make up for. This shows that the constant power method needs to have a properly calibrated resample time for the specific conditions in order to approach the RPP tracking error of the curve-fitting method. A flat irradiance with a longer resample time is when the CPG approach is expected to perform the best, since the longer stretches of more constant irradiance are suitably tracked by the constant power. The next irradiance profiles will be the plunge profile from Figure 26 and 27.

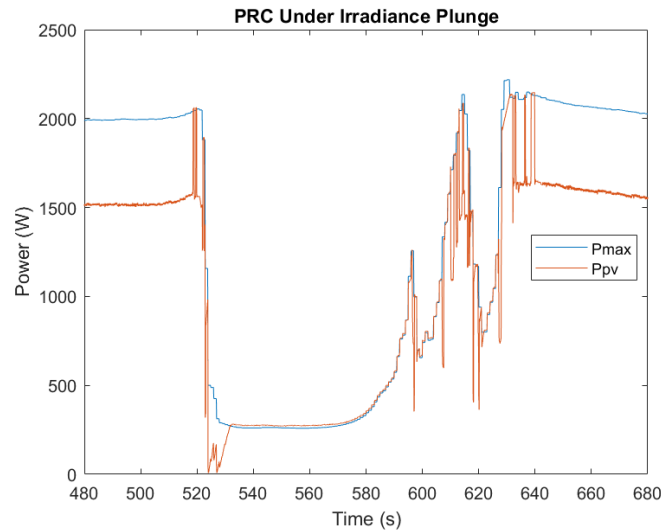


Figure 39: Proposed PRC algorithm under plunge irradiance profile

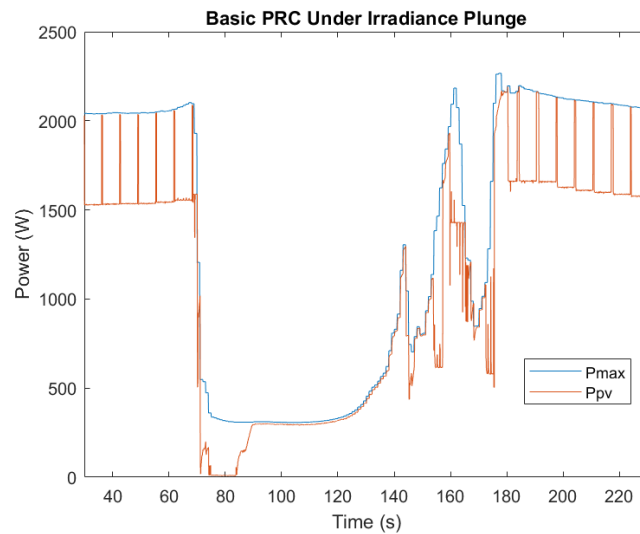


Figure 40: CPG PRC algorithm under plunge irradiance profile

The results of the plunging irradiance test show some discrepancies compared to the simulation. Where the simulation plots showed excellent RPPT in the sampled regions of rapidly fluctuating irradiance, the experimental algorithm struggles to fit new samples into the array in such conditions and therefore is unable to fit a curve to the reserve power. This phenomenon is shown in Figure 41, where the highest index of the sample array is shown to fluctuate and not increase. This happens because the sorting routine will expel points that prevent a curve from being drawn, but if the newest point is the offender then it will re-trigger the new point condition. The issue explained in the previous section of the control cycle being unable to keep up with the IV characteristic in this region is also exacerbated by the measurement noise. The CPG PRC does not fare much better, however. Figure 40 shows the constant power controller also struggles with the large irradiance variation, indicating that this issue is likely universal to PRC algorithms that attempt to sample at runtime.

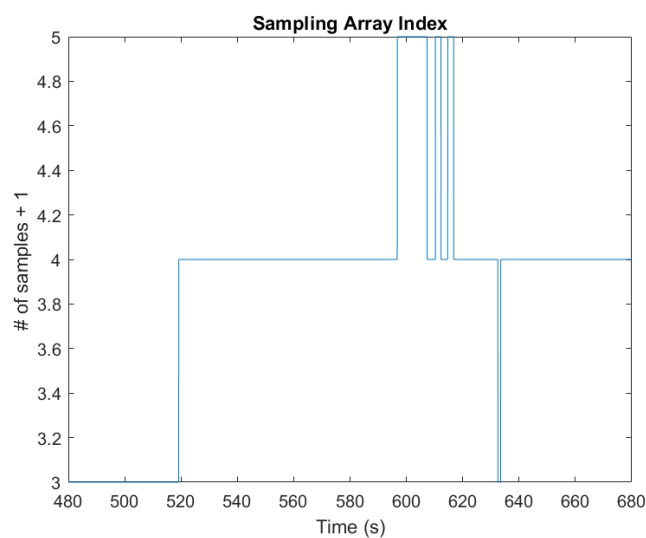


Figure 41: Proposed PRC algorithm sample array index

Also notable is the current collapse after the plunge at around $t = 75\text{s}$. The most likely cause is the operating voltage becoming infeasible when the irradiance drops below a certain point. The PV emulator models particularly large drops in voltage as the irradiance decreases, as shown in Figure 34, so if the algorithm is at a high voltage and increasing it while searching for an RPP, the irradiance plunge could cause it to fall off the IV curve entirely. This is a discrepancy from the simulation, as the low irradiance condition triggers before this happens. Both algorithms manage to recover to the MPP afterwards regardless.

Next, a milder dip profile is examined to see the performance in less extreme variation conditions. The reserve power line is shown in these plots for clarity. This is an improved result from the plunge, but it is still clear that fast-dropping irradiance makes it much more difficult to sample. There is some indication of the dip tracking behavior seen in the simulation, however, particularly at $t = 980\text{s}$ when the operate mode successfully tracks the reserve power on the rebound from the irradiance dip. The error calculations are also much worse than the simulation at 6.21% and 7.51% for the proposed algorithm and the CPG one, respectively. The difficulties of the constant power controller are also quite apparent in the milder dips. Especially after $t = 460\text{s}$ in Figure 43 the controller is stuck at a high power while the irradiance drops, meaning the reserve power at this point is much less than the targeted amount.

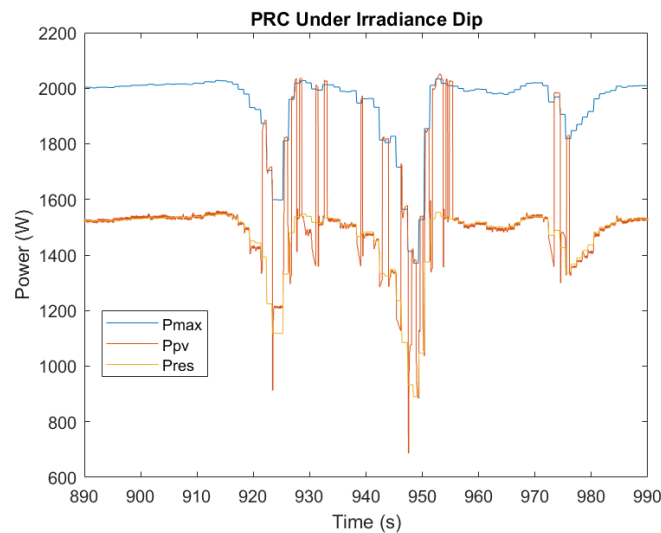


Figure 42: Proposed PRC algorithm under dip irradiance profile

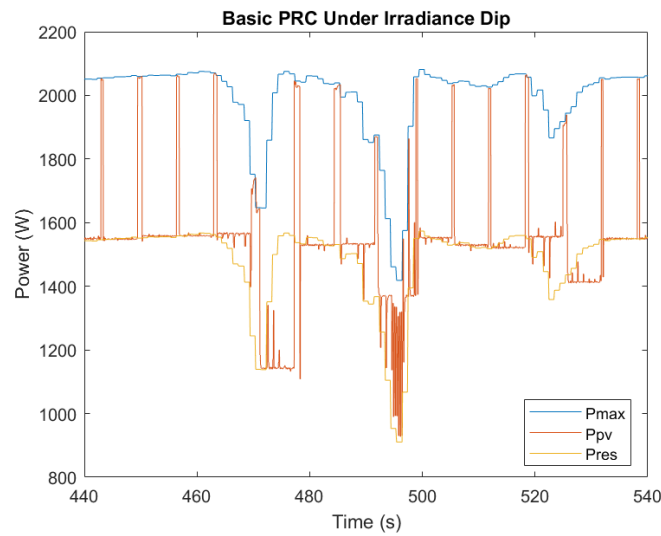


Figure 43: CPG PRC algorithm under dip irradiance profile

Profile	Proposed PRC error	CPG PRC error
Sunny Ramp	1.37%	2.68%
Plunge	N\A	N\A
Dip	6.21%	7.51%

Table 5: Experimental RPPT error results

Overall, the results of the curve-fitting PRC algorithm's performance are promising. The curve-fitting operation mode shows most of the tracking accuracy as predicted by the simulation in relatively calm conditions. The sampling mode was much less effective in irradiance fluctuation than expected, most likely because the extra noise introduced in hardware experiments reduce the quality of the samples and thereby make it difficult to fit a curve that satisfies the constraints. The regions of the most extreme irradiation changes in the plunge profile did not even make it to the operation mode. However, compared to the CPG PRC algorithm, the proposed curve-fitting method performed quite favorably. Its error was lower in every profile tested, and the resampling time constant of the CPG algorithm needed to be tuned to the specific situation in order to achieve comparable behavior. The results therefore show that a quadratic curve-fitting approach is viable for RPP tracking, especially in calm irradiance conditions.

6 Auxiliary Evaluations

6.1 Budget

The scope of the budget for this project is limited as it was carried out in an academic setting with pre-existing borrowed equipment. The only costs considered therefore will be the personal costs of the author, including the personally used software license, as well as the electricity costs for the power system testing.

The author's Erasmus Mundus scholarship payment over a period of five months equals €7000. The laboratory software and equipment are all shared by the lab at the host university and were budgeted for separately by the institution. The only personal software used was MATLAB/Simulink, for which annual licenses cost €290, so the proportional cost for the 5 month work period is €120.83 [33].

Finally, the hardware testing of the PV system and the simulations beforehand used a significant amount of electricity. The average power of the converter during experiments is estimated to be 1kW given the maximum power of the tested array is 2.13kW and most experiments were performed at curtailed power below maximum irradiance. Given the eight week experiment period in 1, only four involved full-system power testing. Assuming 10 hours per week over this period, that gives a testbed energy usage of 40kWh. The author's personal computer ran for an estimated 42 hours per week for 19 weeks at an average power draw of 30W [34] gives 23.94kWh of energy usage from the personal computer for coding and simulations. The industrial rate for electricity in Xi'an, China is 0.58 RMB or €0.07 per kWh [35]. The sum of the electricity cost, scholarship cost, and MATLAB license cost gives the total project budget in Equation 26.

$$(40 + 23.94)(0.07) + 7000 + 120.83 = €7125.1 \quad (26)$$

6.2 Sustainability & Equality

The environmental impact of the project was limited to electricity usage. The author commuted on foot to the lab every day and no new equipment needed to be purchased. There are 530.6 grams of carbon dioxide emissions per kilowatt-hour of electricity produced in China as of the most recent estimate [36], so using the total project energy consumption calculated in the previous section, the total emissions of this project are given as follows:

$$63.94 * 0.5306 = 33.93 \frac{\text{kg}}{\text{CO}_2} \quad (27)$$

For reference, the same amount of carbon is emitted by a 2025 Dacia Sandero TCe 90 driving 282km, which uses less than half of a tank's worth of fuel [37]. It is worth noting that almost the entire field of renewable energy research is undertaken with the ultimate goal of mitigating the emissions of the electricity grid. One of the main rate limiters of more renewable deployment is their adverse effect on grid stability due to lacking inertia. PRC and other technologies enable RES to provide virtual inertia and replace fossil-fuel powered synchronous machines, which will have a much larger effect on emissions than any single research project. This work contributes to the sustainability goals of clean energy and climate action by supporting the knowledge base that will contribute to eliminating fossil electricity.

The social equality implications of this project are minimal at most. The author and both supervisors are men as well as the overwhelming majority of authors cited. It is a well-known issue

that the field of engineering is largely composed of men. It is the responsibility of the author and all men in engineering to confront personal and cultural biases that are hostile to women and discriminate against their contributions to engineering.

7 Conclusions

This work has elaborated a novel proposal for a solar PRC algorithm based on sampling and quadratic curve fitting. First, a conceptual review of VSGs as the primary application of PRC was done to situate the proposal within the greater realm of renewables research. Next was a discussion of the different tactics for implementing PRC specifically for PV systems to give context for the domain of investigation. This also showed the gaps in the existing strategies and what issues the PRC proposal aims to address. The implementation of the algorithm was then detailed, describing the MPPT mode, sampling mode, and operation mode. MPPT uses an established technique to locate MPPs, operate at low irradiance and deploy the reserve power. The sampling mode uses the stored MPP to find a RPP and store it, while ensuring the proper sequentiality and spacing between the samples for a quality curve fit of RPPs across different irradiances. The last mode calculates the curve based on the location within the sampled region and tracks the curve, maintaining a constant reserve power through irradiance changes and minimal sampling.

The next sections discussed the design of an experiment to verify the proposed PRC algorithm. The PV operating point is controlled by a boost converter between the PV and the DC link. The section detailed the converter and controller design according to the testing scenario. Different real-life irradiance profiles were input to the PV model to verify the algorithm's performance in different environmental conditions to test its goal of accurate RPP tracking. The performance was compared to simpler CPG PRC algorithm, inspired by a previous work and employed in the main reference paper for this project, that samples periodically for a RPP and maintains that constant power. The proposed algorithm showed generally showed the capacity for very accurate RPPT in the operation mode, with some indication that rapid irradiance changes would be difficult to sample through. However, the proposed PRC reserve power tracking was up to 10 times more accurate than the CPG strategy in the simulation, depending on profile and parameters. The hardware experiment results confirmed that rapid irradiance changes can prevent a curve from being fitted due to several factors, including the imperfect quadratic relationship between RPPs, measurement noise, and the limitations of the control cycle speed. The algorithm still showed a modest accuracy improvement over the CPG PRC model, which produced significant errors under irradiance changes and requires different tuning for calm and fluctuating conditions. The objectives of this project have therefore been met. The original PRC proposal was developed into a workable algorithm that was proven to be viable in a simulation and test-bench experiment.

7.1 Future Work

This work has raised several avenues for further investigation. Principally, the evaluation of the curve-fitting PRC would be made more complete by integrating it with a VSG inverter and an AC grid, so the purpose of PRC itself could be made more clear by demonstrating its contribution to creating virtual inertia. It would also allow for contrasting the effects on inertia response when supported by constant versus fluctuating power reserves. The experiments performed here also limited the investigated periods to 3-17 minutes in duration, so additional experiments could explore the operation over a full day. This could significantly mitigate some of the issues caused by irradiance fluctuation when the irradiance drops into regions sampled earlier in the day. The algorithm could also be tested in more significant temperature fluctuations. The temperature profiles used in this work were taken during a summer day in Hawaii at a low fidelity, so the temperature only changed by a few degrees and was essentially negligible.

In terms of improving the algorithm, the principle issue to address is the fast irradiance changes. A different tactic for fitting the quadratic curve could be based on a regression of all of the samples rather than a perfect fit to just three. This would eliminate the rigid restrictions for accepting samples and could improve the performance in fluctuating conditions. The tradeoff between samples and accuracy could be more thoroughly explored in this way. Additionally, there are a large number of parameters relevant to the algorithm operation such as averaging time and resampling thresholds that could be experimented with further to find areas of improvement. During the sampling standby, the constant voltage proved a reasonable approximation for the RPP curve over a short range. Future improvements to the algorithm could explore using this feature to avoid futile sampling during the irradiance spikes. The algorithm's performance was also only compared to an accepted but simple alternative. There are several more sophisticated MPP estimation algorithms that could be deployed to compare the PRC against another modern technique.

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A Algorithm Inputs

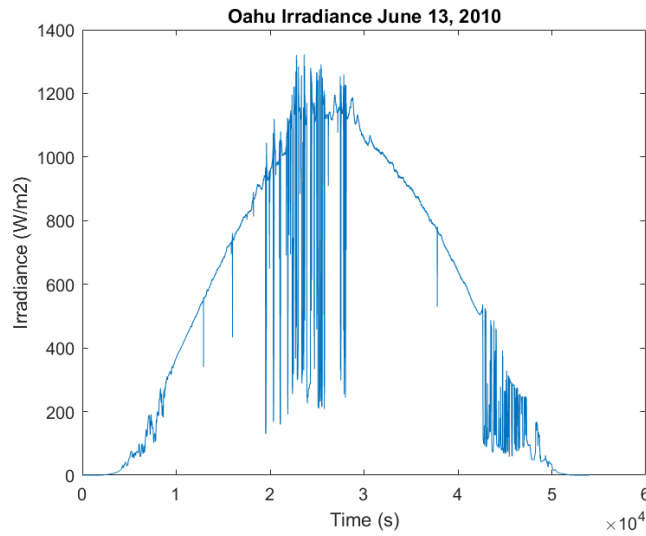


Figure 44: Sample day from [3] with irradiance profiles used

Parameter Name	Value	Note
REPEAT_THRESH_P	30	Minimum power between samples
MPP_SLOPE	1	Minimum slope to arrive at MPP
NEW_POINT	80	Power away from closest sample that curve fit is valid
RESAMPLE_THRESH	45	Power threshold to standby for in sampling mode
RP_THRESH	5	Power threshold to arrive at RPP
RECOVER_FRAC	1.05	Fraction of V_{mp} to jump to from the wrong side of PV curve
MIN_IDX	3	Minimum samples to fit a curve
DELTA_CURVE	0.1	Change in V_{ref} per control cycle in operate mode
LOW_IRR_FRAC	1	Minimum fraction of the RPP / reserve power
RESAMPLE_CYCLES	60	Number of control cycles to hold for Basic PRC
RPP_KP	1/34	Proportional power error term to sample RPP

Table 6: Simulation constants

The simulation constants are mostly derived through experiment, and are dependent on the PV size. The most important are the power thresholds, which will impact accuracy if tuned too small.

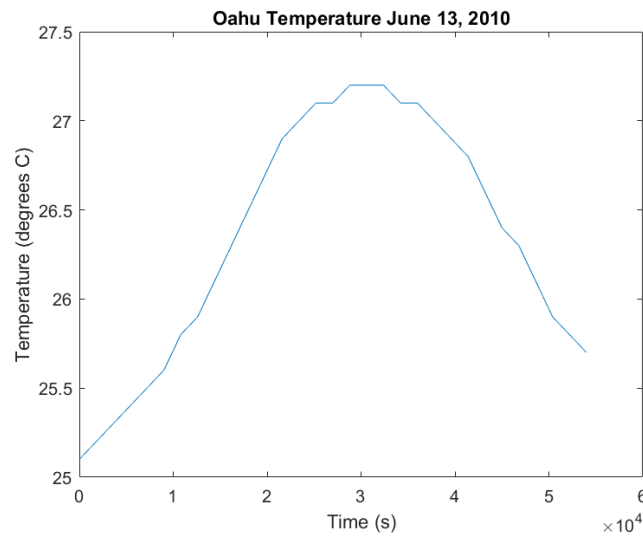


Figure 45: Temperature profile of example day

The temperature data available for the location of the irradiance was at a 30 minute interval. The temperature profiles corresponding to each irradiance profile was applied in simulation and experiment but had no discernible effect.

B Algorithm Outputs

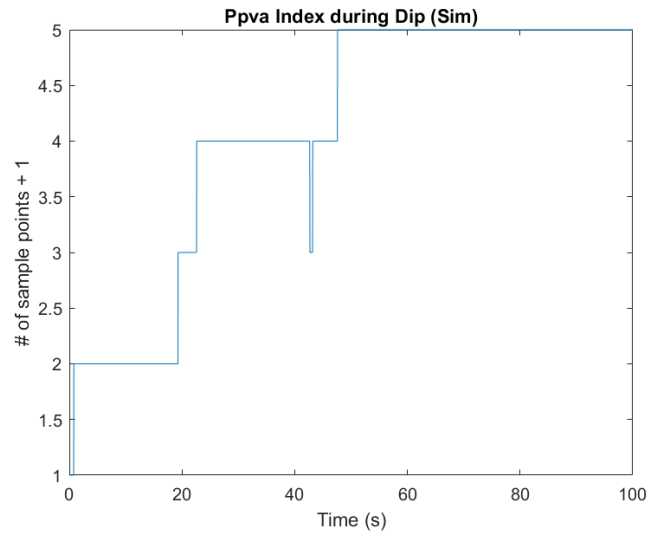


Figure 46: Array index from the dip profile simulation

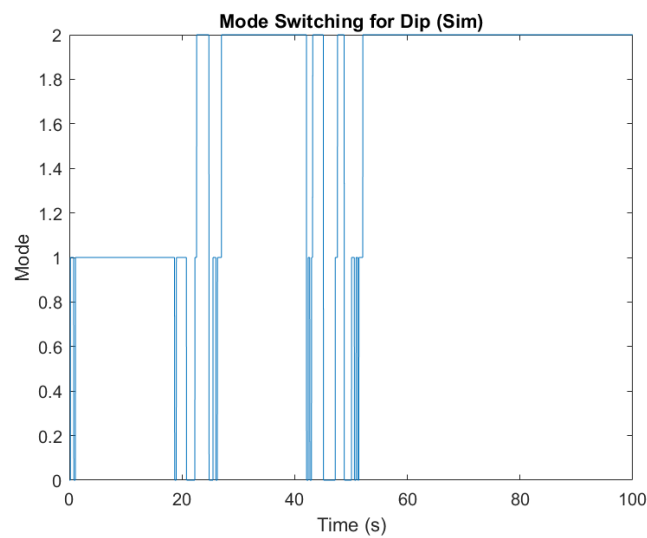


Figure 47: Mode graph from the dip profile simulation

Figures 46 and 47 show representative profiles of the internal algorithm variables for the number of samples stored and the current mode. Both are mentioned or included in the results sections where directly relevant to a notable behavior, such as operation mode tracking or sampling failures.

C Code

The mode function C implementations and subroutines are shown below:

```
void mppt_mode(float Vpv, float Ppv, float dV, float dP){
    if(s_MPPf){ //return to previous MPP
        s_Vref = s_Vmpp;
        s_MPPf = 0;
        return;
    } else {s_Vref = s_Vrefprev;}
    if(!s_Pprev){return;} //wait for power measurement

    //MAIN P&O Logic
    if(Ppv > s_Pprev){
        if(s_MPPT_dir){
            s_Vref = s_Vrefprev + s_deltaVref;
        }
        else {
            s_Vref = s_Vrefprev - s_deltaVref;
        }
    } else {
        if(s_MPPT_dir){
            s_Vref = s_Vrefprev - s_deltaVref;
            s_MPPT_dir = 0;
        }
        else {
            s_Vref = s_Vrefprev + s_deltaVref;
            s_MPPT_dir = 1;
        }
    }

    //Check for max power
    float PVslope = mwabs(dP/dV);
    if(PVslope < MPP_SLOPE){
        s_Pmax = Ppv;
        s_Vmpp = Vpv;
        s_MPPf = 1;
        s_mode = 1;
        s_Vref = s_Vrefprev;
        if(Ppv < s_Preserve*(2*LOW_IRR_FRAC)){ //if low irradiance, stay in MPPT
            s_MPPf = 0;
            s_mode = 0;
        }
    }
}

void prc_measure_mode(float Vpv, float Ppv, float dV, float dP){
    if(s_Vpvmpre && !s_standby){ //return to last known point
        s_Vref = s_Vpvmpre;
    }
}
```

```

    s_Vpvmppre = 0.0f;
    return;
}
if ((Ppv < s_Preserve*LOW_IRR_FRAC)) { //low irradiance check
    s_Vpvmppre = 0.0f;
    s_standby = 0;
    s_mode = 0;
    return;
}
    if(mwabs(Ppv - s_Ppvmppre) > RESAMPLE_THRESH){ //prevent mode 0-1 oscillation
        s_standby2 = 0;
    }

//search for reserve power
float target = s_Pmax - s_Preserve;
updateDeltaVprc(Ppv);
if(Vpv < s_Vmpp){
    s_Vref = RECOVER_FRAC*s_Vmpp;
    return;}
else {s_Vref = s_Vrefprev + s_deltaVprc;}

bool onTarget = mwabs(Ppv - target) <= RP_THRESH;
test_float = min_abs_diff(Ppv,s_Ppva,s_k);
bool noRepeatP = test_float >= REPEAT_THRESH_P;
bool rightSide = Vpv > s_Vmpp;

if(s_fine && onTarget && !s_standby){
    s_Vpvmppre = Vpv;
    s_standby = 1;
    if(onTarget && rightSide && noRepeatP){
        int idxR = insertPoint(Vpv,Ppv);
        if(s_k >= MIN_IDX && !s_sortFail){
            s_mode = 2;
            s_standby = 0;
            s_standby2 = 0;
        } else if(s_sortFail != 0){
            s_test += 1;
            if(s_k ==1){
                s_Vpva[0] = Vpv;
                s_Ppva[0] = Ppv;
            }else{
                int i = idxR;
                for( i =idxR; i<(s_k); i++){ //remove point causing sortfail
                    s_Vpva[i] = s_Vpva[i+1];
                    s_Ppva[i] = s_Ppva[i+1];
                }
                s_k -= 1;
            }
        }
    }
}

```

```

        s_sortFail = 0;
        s_mode = 0;
        s_standby = 0;
        s_standby2 = 0;
    }
}
s_fine = 0;
}

// Exit conditions
if(s_standby && !s_standby2 && s_k>=MIN_IDX && Ppv < s_Ppva[s_k-1] && Ppv > s_Ppva[0]){
    s_mode = 2;
    s_standby = 0;
    s_standby2 = 0;
} else if(s_standby && noRepeatP && !s_standby2){
    s_mode = 0;
    s_standby = 0;
    s_standby2 = 1;
    s_Ppvmpr = Ppv;
}
}

float min_abs_diff(float value, const float* array, int array_size) {
    if (!array_size) {return value;}

    float min_diff = mwabs(value - array[0]);
    int i = 1;
    for (i = 1; i < array_size; i++) {
        float diff = mwabs(value - array[i]);
        if (diff < min_diff) {min_diff = diff;}
    }

    return min_diff;
}

void updateDeltaVprc(float Ppv){
    // s_err = mwabs(Ppv - (s_Pmax - s_Preserve));
    s_err = Ppv - (s_Pmax - s_Preserve);
    s_errInt += s_err;
    //controller to search for PRC point
    // Kp = 1/23 for 600, 1/34 for 1000
    s_deltaVprc = RPP_KP*s_err;

    if(mwabs(s_err/(s_Pmax- s_Preserve)) < 0.025){s_fine = 1;}
    if(s_standby){s_deltaVprc = 0.0f;}
}

int insertPoint(float Vpv, float Ppv){

```

```

int idxP = s_k;
int idxV = s_k;
int idxR = 0;
int i = 0;
for(i=0; i<(s_k); i++){
    if(Ppv < s_Ppva[i]){ //ascending
        idxP = i;
        break;
    }
}
for(i=0; i<(s_k); i++){
    if(Vpv < s_Vpva[i]){ //ascending
        idxV = i;
        break;
    }
}

if(idxP != idxV){
    s_sortFail = 1;
    idxR = idxV; //remove the voltage on failure
    return idxR;
}

if(s_k){
    for(i = (s_k-1); i >= idxP; i--){
        s_Ppva[i+1] = s_Ppva[i];
        s_Vpva[i+1] = s_Vpva[i];
    }
}

s_Ppva[idxP] = Ppv;
s_Vpva[idxV] = Vpv;
s_k += 1;
s_sortFail = 0;
return idxR;
}

void prc_operate_mode(float Vpv, float Ppv){
    int idx[3];
    if(Ppv < s_Ppva[2] || s_k == MIN_IDX){
        idx[0] = 0;
        idx[1] = 1;
        idx[2] = 2;
    }else if(Ppv > s_Ppva[s_k-2]){
        idx[0] = s_k - 3;
        idx[1] = s_k - 2;
        idx[2] = s_k - 1;
    }else{
        int minIdx = 1;

```

```

float minDiff = mwabs(Ppv - s_Ppva[minIdx]);
int i = 1;
for(i=1; i<(s_k-1); i++){
    if(mwabs(Ppv - s_Ppva[i]) < minDiff){
        minDiff = mwabs(Ppv - s_Ppva[i]);
        minIdx = i;    //search for the smallest difference among sample points
    }
}
idx[0] = minIdx - 1;
idx[1] = minIdx;      //center bracket around that point
idx[2] = minIdx + 1;
}

//get all the curve fit points
s_Vpm[0] = s_Vpva[idx[0]];
s_Vpm[1] = s_Vpva[idx[1]];
s_Vpm[2] = s_Vpva[idx[2]];
s_Ppm[0] = s_Ppva[idx[0]];
s_Ppm[1] = s_Ppva[idx[1]];
s_Ppm[2] = s_Ppva[idx[2]];

double dV21 = (double) (s_Vpm[1] - s_Vpm[0]);
double dV32 = (double) (s_Vpm[2] - s_Vpm[1]);
double dV13 = (double) (s_Vpm[0] - s_Vpm[2]);
double dP21 = (double) (s_Ppm[1] - s_Ppm[0]);
double dP32 = (double) (s_Ppm[2] - s_Ppm[1]);
double P0 = (double) s_Ppm[0];
double V0 = (double) s_Vpm[0];

s_A = (dP21*dV32 - dP32*dV21) / (dV21 * dV32 * dV13);
s_B = (dP21 - s_A*(s_Vpm[1]*s_Vpm[1] - s_Vpm[0]*s_Vpm[0])) / dV21;
s_C = P0 - s_A*V0*V0 - s_B*V0;

double Vpvd = (double) Vpv;
double Pcurve = (s_A*Vpvd*Vpvd + s_B*Vpvd + s_C);
double Vcrit = -s_B/(2*s_A);
double Pcrit = s_A*Vcrit*Vcrit + s_B*Vcrit + s_C;

bool fallOff = (s_A < 0 && (double)Ppv > Pcrit) || (s_A > 0 && (double)Ppv < Pcrit);
int idxR = -1;
if(fallOff){
    if(s_A < 0){idxR = s_k - 1;}
    else{idxR = 0;}
    int i = idxR;
    for( i = idxR; i<(s_k); i++){ //remove point causing sortfail
        s_Vpva[i] = s_Vpva[i+1];
        s_Ppva[i] = s_Ppva[i+1];
    }
}

```

```

    s_k -= 1;
}

if( (double)Ppv <= Pcurve){
    s_Vref = s_Vrefprev - DELTA_CURVE;
} else {
    s_Vref = s_Vrefprev + DELTA_CURVE;
}

bool lowIrr = Ppv < s_Preserve*LOW_IRR_FRAC;

// exit conditions
test_float = min_abs_diff(Ppv,s_Ppva,s_k);
if(test_float >= NEW_POINT || fallOff || lowIrr || s_k < MIN_IDX){
    s_Vpvmpr = Vpv;
    s_mode = 0;
    s_standby2 = 0;
}
}

```

The main function is shown below:

```

if (modepv==0.0f) { // controlled by switchboard
    count+=1.0f;
    if (count>1000.0f)
    {
        sumv+=vpv;
        sumi+=ipv;
    }
    if (count == 2000.0f) //0.1s
    {
        vpvmean=sumv*10e-4f;
        ipvmean=sumi*10e-4f;
        ppv=vpvmean*ipvmean;
        //lpf( ppv, &ppv_f, b1_lpf_t200, a0_lpf_t200 );
        dv=vpvmean-vpvmean_pre;
        // //dp=ppv_f-ppv_pre;
        dp=ppv-ppv_pre;
        vpvmean_pre=vpvmean;
        ppv_pre=ppv;
        // s_mode = 0; // lock in MPPT
        switch (s_mode){
            case 0:
                mppt_mode(vpvmean,ppv,dv,dp);
                break;
            case 1:
                prc_measure_mode(vpvmean,ppv,dv,dp);
                break;

```

```

    case 2:
        prc_operate_mode(vpvmean,ppv);
        break;
    }
    vpvref = s_Vref;
    s_Vrefprev = s_Vref;
    s_Pprev = ppv;
    s_Vprev = vpvmean;
    // vpvref = 310.0f;
    if((di_switch & SW_PRO_ON) == SW_PRO_ON){//switch 9 for step response
        s_mode = 0;
    }
    slope=dp/dv;
    count=0.0f;
    sumv=0.0f;
    sumi=0.0f;
}
vpv_err = -vpvref + vpv;
// single stage ctrl
// pi_ctrl( vpv_err, &vpv_err_pre, &d_pv, 1.0f, 800.0f, 0.99f, 0.01f );
//inner loop
pi_ctrl(vpv_err, &vpv_err_pre, &Ipvref, a0_pvctrl, a1_pvctrl, 10.0f, 0.0f );
// Ipvref = 4.0f;
ipv_err = Ipvref - ipv ; //12.0f, -2.0f
//inner loop
pi_ctrl( ipv_err, &ipv_err_pre, &d_pv, a0_ipvctrl, a1_ipvctrl, 0.99f, 0.01f );
// d_pv = 0.999f; // dpv is duty cycle of NEGATIVE transistor
//a0 = kp, a1 = ki //20 22 16 14 12 0
IPFPGA_write(BDN1, 0x24, (d_pv)*Carrier_count);/* BOOST (16bit) */
}

```

D Testbench Pictures

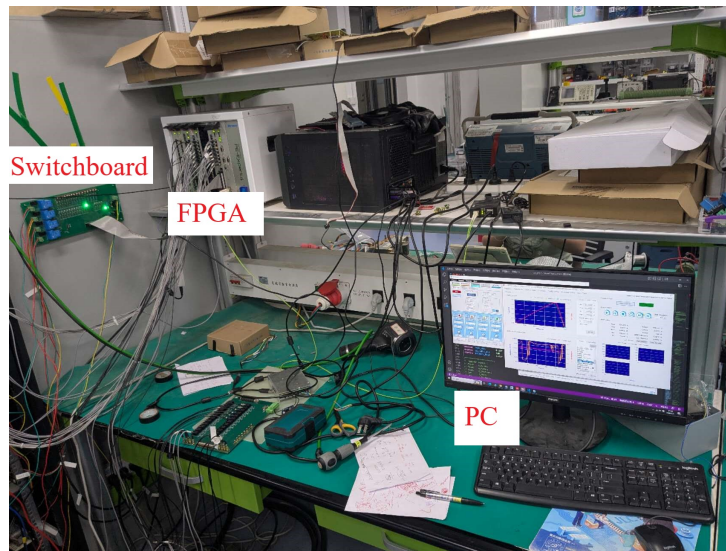


Figure 48: MywayPE4 FPGA system with accompanying switchboard

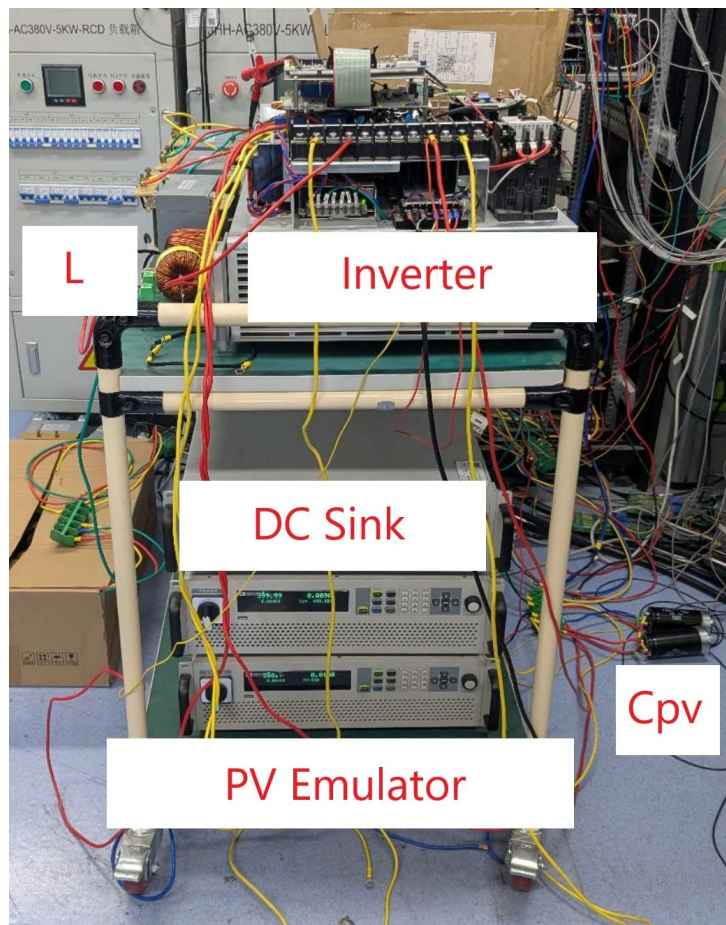


Figure 49: Experimental circuit setup